

72nd BPSC PRELIMS

प्रहार

Test Series

Test-21/5

FULL LENGTH TEST

Solution

English



BPSC TEST-21/5 (SOLUTION)

Q 1. Ans. B

Explanation:

- According to the Watson–Crick model of DNA (1953):
 - ✓ DNA consists of two polynucleotide strands twisted around each other to form a double helix.
 - ✓ The two strands run in opposite directions, hence they are described as anti-parallel.
 - ✓ One strand runs in the $5' \rightarrow 3'$ direction, while the complementary strand runs in the $3' \rightarrow 5'$ direction.
 - ✓ Nitrogenous bases pair specifically through hydrogen bonds:
 - Adenine (A) $\leftrightarrow$ Thymine (T) (2 hydrogen bonds)
 - Guanine (G) $\leftrightarrow$ Cytosine (C) (3 hydrogen bonds)

Q 2. Ans. B

Explanation:

- According to the Impulse-Momentum Theorem, the change in momentum of a body is equal to the impulse applied to it.
- Impulse = Force $\times$ Time
- Given:
 - ✓ Force (F) = 6 N
 - ✓ Time (t) = 2 s
 - ✓ Initial momentum = 0 (since the body is initially at rest)
- Therefore, Change in momentum = $6 \times 2 = 12$ kg·m/s
- Since the initial momentum is zero, the final momentum of the body is:
- Final momentum = $0 + 12 = 12$ kg·m/s

Q 3. Ans. C

Explanation:

In uniform circular motion, a particle moves along a circular path with a constant speed, but its direction of motion changes continuously.

- **Statement 1 is correct:** In uniform circular motion, a particle moves along a circular path with a constant magnitude of velocity, which is called speed. The term “uniform” itself indicates that the particle covers equal distances in equal intervals of time. Therefore, while moving around the circle, the speed of the particle does not change, even though its direction continuously changes.

- **Statement 2 is incorrect:** Velocity is a vector quantity that depends on both magnitude and direction. Although the speed of a particle in uniform circular motion remains constant, its direction changes at every point on the circular path. Since a change in direction results in a change in velocity, the velocity of the particle cannot remain constant during uniform circular motion.
- **Statement 3 is correct:** A particle moving in a circular path experiences centripetal acceleration, which is responsible for continuously changing the direction of its velocity. This acceleration always acts towards the centre of the circle. Hence, the acceleration is always directed towards the centre.

Q 4. Ans. A

Explanation:

- In a freely falling lift, effective gravity becomes zero, so the pendulum does not oscillate and its frequency becomes zero. The restoring force responsible for the oscillation of a simple pendulum is provided by gravity. When a lift falls freely under gravity, both the support of the pendulum and the bob accelerate downward with the same acceleration ‘g’. As a result, the bob experiences a state of apparent weightlessness, and the effective gravitational acceleration inside the lift becomes zero. Since the restoring force is proportional to the effective gravity, no restoring force acts on the bob to bring it back to its mean position when displaced.

Q 5. Ans. A

Explanation:

- **Statement 1 is correct:** Surface tension is defined as the tangential force acting per unit length on either side of an imaginary line drawn on the free surface of a liquid. This force acts in the plane of the liquid surface and perpendicular to the imaginary line. Due to this property, the liquid surface behaves like a stretched elastic membrane and tends to minimise its surface area.
- **Statement 2 is correct:** Surface tension arises due to cohesive forces between liquid molecules. As the temperature increases, the kinetic energy of the molecules increases, weakening the intermolecular attraction. Consequently, the surface tension decreases with increasing temperature and becomes nearly zero at the critical temperature of the liquid.

- **Statement 3 is incorrect:** Surface tension is **not a vector quantity**. Although it is defined in terms of force per unit length, surface tension itself is considered a **scalar quantity** because it has only magnitude at a given liquid interface and temperature. The direction is associated with the force produced by surface tension, not with the surface tension coefficient itself.

Q 6. Ans. C

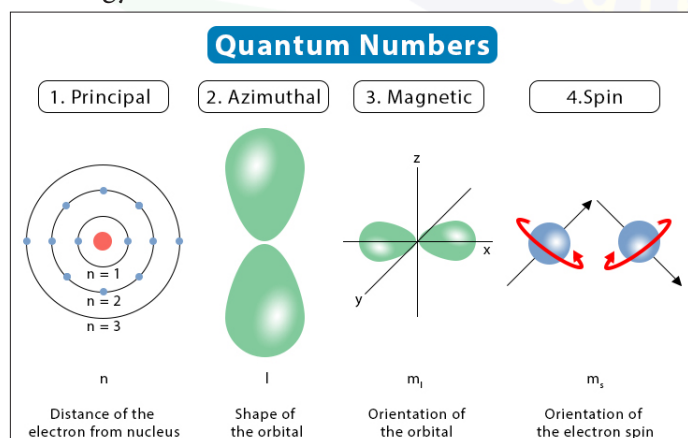
Explanation:

- When a metal rod is heated, the atoms of the metal gain kinetic energy and vibrate more vigorously about their mean positions. Due to the asymmetric nature of the interatomic force-distance curve, the average separation between neighbouring atoms increases. As a result, the dimensions of the metal rod increase, leading to thermal expansion. Thermal expansion occurs because the **mean separation between atoms increases when the amplitude of atomic vibrations increases with temperature**. The atoms remain bound to their equilibrium positions but oscillate with larger amplitudes, leading to an increase in the dimensions of the solid.

Q 7. Ans. A

Explanation:

- **Statement 1 is correct:** The principal quantum number (n) identifies the main shell or energy level in which an electron is present. It has values 1, 2, 3, 4, and so on, corresponding to the K, L, M, N shells, respectively. As the value of n increases, the average distance of the electron from the nucleus increases, leading to a larger orbital size. Thus, the principal quantum number determines both the size and the energy of the orbital.



- **Statement 2 is correct:** Magnetic quantum number (m_l) specifies the orientation of an orbital in space. For a given value of the azimuthal quantum number

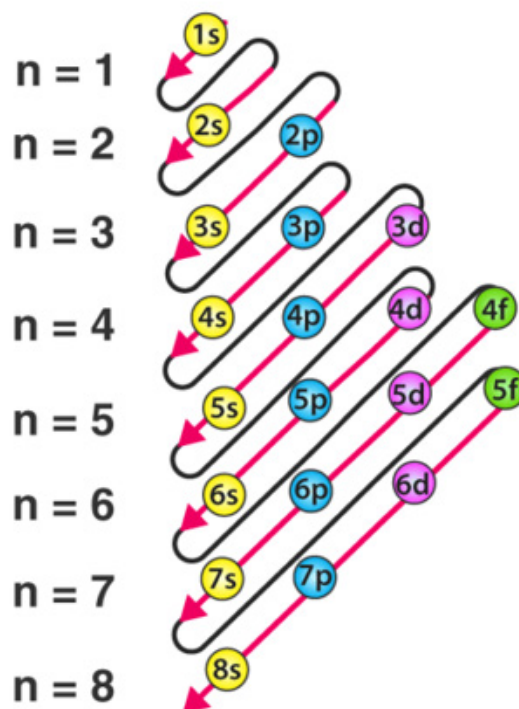
(l), the magnetic quantum number can have values ranging from $-l$ to $+l$, including zero. Therefore, the magnetic quantum number determines the orientation of an orbital in space.

- **Statement 3 is incorrect:** The azimuthal quantum number (l) can take integral values from 0 to $(n - 1)$ for a given principal quantum number n . Therefore, the total number of possible values of l is n , not $(n + 1)$. For example, when $n = 3$, the possible values of l are 0, 1, and 2, giving three values in total. Similarly, for $n = 4$, the possible values are 0, 1, 2, and 3, giving four values. Hence, the statement is incorrect because the number of possible values of l is equal to n .

Q 8. Ans. D

Explanation:

- The relative energies of orbitals in multi-electron atoms are determined by the **($n + l$) rule**. For the 4d orbital, the principal quantum number (n) is 4, and the azimuthal quantum number (l) is 2. Therefore, $n + l = 4 + 2 = 6$, which is the highest among the given orbitals (4s, 3d, 4p, and 4d). Since orbitals with higher values of $(n + l)$ possess higher energy, the 4d orbital has the highest energy. Aufbau filling order in which 4d orbitals are filled only after lower-energy orbitals, such as 5s, indicating their relatively high energy. Hence, **4d is the highest-energy orbital among the given options**.



Q 9. Ans. B**Explanation:**

- Molecular orbital (MO) theory explains that oxygen contains **two unpaired electrons** in its antibonding π^* (2p) molecular orbitals. The electronic configuration of O_2 in terms of molecular orbitals leaves one electron in each of the two degenerate π^* orbitals, resulting in the presence of unpaired electrons. Since substances containing one or more unpaired electrons are **paramagnetic**, O_2 is paramagnetic. **This experimental observation could not be explained by Lewis structures but is successfully explained by molecular orbital theory**, making it one of the strongest pieces of evidence supporting MO theory.
- N_2 , F_2 , and Cl_2 have all their electrons paired in molecular orbitals and are therefore **diamagnetic**, whereas O_2 is **paramagnetic due to two unpaired electrons**.

Q 10. Ans. A**Explanation:**

- The standard enthalpy of formation of CO_2 is the enthalpy change when **1 mole of carbon (12 g)** combines completely with oxygen to form **1 mole of CO_2** .
 $\checkmark C(s) + O_2(g) \rightarrow CO_2(g)$
 $\checkmark \Delta H^\circ = -393.5 \text{ kJ mol}^{-1}$
- **This means that the combustion of 12 g of carbon releases 393.5 kJ of heat. Therefore, for 1 g of carbon:**
 $\checkmark \Delta H = (-393.5 \text{ kJ}) / 12$
 $\checkmark \Delta H = -32.8 \text{ kJ}$
- Hence, the enthalpy of combustion of **1 g of carbon is -32.8 kJ**.
- A negative value of enthalpy change indicates an **exothermic reaction**, meaning heat is released to the surroundings. Combustion reactions are typically highly exothermic and therefore have negative enthalpy values.

Q 11. 11. Ans. C**Explanation:**

- The Sangam literature of ancient Tamilakam classified the land into five distinct ecological zones, known as the **Tinai**. Each zone had a specific geographical feature, a corresponding deity, and a characteristic occupation based on the resources available in that environment.
- **Neithal** is the coastal landscape. Because of its

proximity to the sea, the primary occupations of the inhabitants (known as *Parathavar*) were **fishing, salt manufacturing, and seafaring/maritime trade**.

Tinai	Geographical Feature	Primary Occupation
Kurinji	Hilly/Mountainous	Hunting and gathering
Mullai	Forest/Pastoral	Cattle rearing and dairying
Marudam	Fertile Agricultural plains	Farming (Paddy cultivation)
Palai	Arid/Desert	Heroic exploits and robbery (due to scarcity)
Neithal	Coastal	Fishing and salt production

Q 12. Ans. A**Explanation:**

- **Statement 1 is correct:** The Harappan people developed a highly sophisticated and standardized system of weights and measures. These weights were generally made of a stone called **chert** and were usually cubical with no markings. Interestingly, the lower denominations of these weights followed a binary system (1, 2, 4, 8, 16, 32, 64, and so on up to 12,800), while the higher denominations followed the decimal system. The ratio of 16 (where 16 units equaled 1) was indeed a foundational part of their calculation system.
- **Statement 2 is incorrect:** The Harappan script remains one of the great "undeciphered" mysteries of the ancient world. While archaeologists have found many seals and inscriptions, there is no "Rosetta Stone" equivalent for the Harappan script. No bilingual inscriptions (an inscription written in both the Harappan language and a known, readable script like Sumerian or Akkadian) have ever been discovered in Mesopotamia or elsewhere. Because of this, despite numerous attempts by scholars using advanced computational methods, the script remains largely unintelligible today.

Q 13. Ans. B**Explanation:**

- **Statement 1 is correct:** The Early Vedic economy was primarily pastoral. Cattle were the most important form of wealth, and society revolved around them. The term *Gavisthi* (literally "searching for cows") was frequently used in the Rig Veda to

describe conflicts or wars, highlighting that battles were often fought to capture cattle.

- **Statement 2 is incorrect:** This statement describes the opposite of what occurred during the transition to the Later Vedic period. In the Early Vedic period, the *Sabha* (an assembly of elders) and *Samiti* (a general tribal assembly) held significant power and exerted control over the king. However, during the Later Vedic period, the power of these assemblies declined sharply. As the tribal structure transformed into a more organized territorial state, the king's authority became stronger, hereditary, and more centralized. By the end of this period, the *Sabha* and *Samiti* lost their ability to check the king's power, effectively making the king more influential than the assemblies.

Q 14. Ans. B

Explanation:

- The administrative structure of the Gupta Empire was highly organized and decentralised, operating through a clear hierarchy of territorial divisions.
- The largest administrative unit right below the central imperial government was the province, known as a Bhukti (also sometimes referred to as Desha or Rashtra in certain texts, but Bhukti is the strict administrative term for the provincial level). These provinces were governed by high-ranking officials called Uparikas, who were typically appointed directly by the Gupta king and were often princes or members of the royal family.

Administrative Unit	Modern Equivalent	Title of the Head
Rajya	Empire	Maharajadhiraja
Bhukti	Province	Uparika
Vishaya	District	Vishayapati
Vithi	Sub-district / Group of villages	Vithimahattara
Grama	Village	Gramika

Q 15. Ans. D

Explanation:

- In the highly centralised Mauryan administration detailed in Kautilya's Arthashastra, the **Sannidhata** served as the Chief Treasurer or the Custodian of the State Treasury. This official was responsible for the safe custody of cash, valuables, and the building of storehouses for grains, weapons, and forest produce.

- **Assessment and collection of land revenue:** This was the responsibility of the **Samaharta** (the Collector General), who oversaw the entire revenue collection system across the empire.
- **Supervision of frontier posts:** Frontier forts and outposts were guarded by an official known as the **Antapala**.
- **Administration of civil and criminal courts:** The Mauryan judicial system had two types of courts. Civil courts (**Dharmasthiya**) were presided over by judges called **Dharmasthas**, while criminal courts (**Kantakasodhana**) were presided over by officers known as **Pradeshtas**.

Q 16. Ans. A

Explanation:

- **Lothal**, located in the Bhal region of modern-day Gujarat, is one of the most prominent sites of the Indus Valley Civilisation. It is uniquely famous for the discovery of a massive, rectangular brick-built basin, which archaeologists (notably S. R. Rao) have been identified as the world's earliest known **dockyard**.
- This structure was connected to the ancient course of the Sabarmati River, allowing ships to navigate from the Gulf of Khambhat directly into the city. It served as a major centre for maritime trade, facilitating the export of beads, gems, and valuable ornaments to regions as far as Mesopotamia.
- **Dholavira:** Located in the Rann of Kutch (Gujarat), it is renowned for its highly sophisticated water management system, featuring large stone-built reservoirs and check dams, rather than a maritime harbour.
- **Kalibangan:** Located in Rajasthan, it is famous for yielding evidence of the world's earliest ploughed agricultural field and ritualistic fire altars.
- **Banawali:** Located in Haryana, it is known for the discovery of a high-quality terracotta model of a plough and its unique radial street planning.

Q 17. Ans. B

Explanation:

- **Statement 1 is incorrect:** While Major Rock Edict XIII does indeed provide a vivid account of the Kalinga War and Ashoka's subsequent remorse, it was **not** inscribed in the Brahmi script throughout the empire. Ashoka was highly pragmatic and used local scripts to ensure his subjects could read his messages.
 - ✓ In the eastern, central, and southern parts of

the empire, the edicts were inscribed in **Brahmi** (written in the Prakrit language).

- ✓ However, in the north-western parts of the empire (modern-day Pakistan), edicts like those at Shahbazgarhi and Mansehra were inscribed in the **Kharoshthi** script.
- ✓ Further west in modern-day Afghanistan (Kandahar), his edicts were inscribed in **Greek** and **Aramaic**.

- **Statement 2 is correct:** In the vast majority of his inscriptions, Ashoka does not refer to himself by his actual name, but rather by his royal titles: Devanampiya (Beloved of the Gods) and Piyadasi (He who looks on with affection/One of amiable appearance). Through these edicts, he strongly advocated for his policy of Dhamma, which included a deep respect and tolerance for all religious sects. Major Rock Edict XII, in particular, is an explicit plea for religious tolerance, asking people to respect each other's beliefs and refrain from unnecessarily praising their own sect while criticising others.

Q 18. Ans. A

Explanation:

- The Megalithic culture of South India, which primarily coincides with the Iron Age in the region (c. 1000 BCE to 300 CE), is distinctively associated with **Black and Red Ware (BRW)**.
- This pottery is found abundantly in Megalithic burial sites (such as urn burials, cists, and dolmens) across the southern peninsula. It gets its name from its distinct appearance, a completely black interior and rim, with a red exterior. This dual-colour effect was achieved through a specific technique called "inverted firing," where the pots were placed upside down in the kiln, limiting oxygen flow to the inside (turning it black) while exposing the outside to oxygen (turning it red).
- **Northern Black Polished Ware (NBPW):** This was a highly glossy, luxury pottery associated with the second urbanisation in the Gangetic plains, the rise of the Mahajanapadas, and the Mauryan Empire in North India (c. 6th century BCE to 2nd century BCE).
- **Ochre Coloured Pottery (OCP):** This is a much older pottery type associated with the Chalcolithic (Copper Age) farming communities of the Ganga-Yamuna Doab (c. 2000 BCE to 1500 BCE).
- **Painted Grey Ware (PGW):** This fine, grey pottery with black geometric patterns is predominantly

found in North-Western India and the Gangetic divide, strongly associated with the Later Vedic period and early Iron Age settlements (c. 1200 BCE to 600 BCE).

Q 19. Ans. B

Explanation:

- **The system of Dual Government (1765–1772) was introduced in Bengal by Robert Clive following the Treaty of Allahabad. This system fundamentally separated executive power from administrative responsibility:**
 - ✓ **Diwani Rights:** The East India Company secured the formal right to collect revenues of Bengal, Bihar, and Orissa directly from the Mughal Emperor. This gave the Company complete control over the financial resources of the region.
 - ✓ **Nizamat Functions:** The responsibilities of military defence, police, and the administration of justice (judiciary) remained nominally with the Nawab of Bengal. To exercise this, the Company appointed two Deputy Diwans—Mohammad Reza Khan for Bengal and Raja Shitab Roy for Bihar—who acted on behalf of both the Company and the Nawab.

Q 20. Ans. B

Explanation:

- The Orientalist–Anglicist controversy was a major debate on educational policy in British India during the 1820s and 1830s.
 - ✓ The Orientalists favoured the promotion of traditional Indian learning through classical languages such as Sanskrit, Arabic, and Persian. They argued that government funds should be used to preserve and promote India's indigenous intellectual traditions.
 - ✓ The Anglicists advocated the spread of Western knowledge, science, and literature through the medium of English. They believed that English education would modernise Indian society and provide access to contemporary scientific and intellectual developments.
- The controversy was eventually settled in favour of the Anglicists through Macaulay's Minute and Lord William Bentinck's educational resolution of 1835, which made English the principal medium for higher education and directed government funds towards Western education.

Q 21. Ans. B

Explanation:

- Pitt's India Act of 1784 distinguished between the commercial and political functions of the Company and established a Board of Control to manage the political affairs, while the Court of Directors continued to manage the commercial affairs, thereby setting up a system of double government.
- Pitt's India Act did not abolish the East India Company; the Company was abolished only by the Government of India Act of 1858, enacted after the revolt of 1857. The 1784 Act, in fact, strengthened the control of the British Government over the Company's affairs.
- The Act gave the British Government supreme control over the Company's affairs and its administration in India, and for the first time the Company's territories in India were called the British possessions in India. It was a landmark because it placed Indian affairs under the direct control of the British Government, a control that culminated in the assumption of power by the Crown in 1858.

Q 22. Ans. A

Explanation:

- **Statement 1 is correct:** The Drafting Committee, set up in August 1947, had seven members, namely B. R. Ambedkar, N. Gopalaswami Ayyangar, Alladi Krishnaswami Ayyar, K. M. Munshi, Mohammad Saadulla, N. Madhava Rau, and T. T. Krishnamachari, the last two replacing B. L. Mitter and D. P. Khaitan.
- **Statement 2 is correct:** B. R. Ambedkar was the chairman of the Drafting Committee, which was the most important of the committees of the Constituent Assembly and was entrusted with preparing the draft of the new Constitution.
- **Statement 3 is incorrect:** B. N. Rau was not a member of the Drafting Committee; he was appointed the constitutional advisor to the Constituent Assembly and prepared the initial draft of the Constitution, which the Drafting Committee then scrutinised and revised. The Assembly took two years, eleven months, and eighteen days to complete the Constitution, which was adopted on 26 November 1949.

Q 23. Ans. A

Explanation:

- **Statement 1 is correct:** The Preamble is neither a source of power for the legislature nor a limitation

upon the powers of the legislature; it is non-justiciable, which means that its provisions are not enforceable in a court of law, and it merely indicates the source, objectives, and philosophy of the Constitution.

- **Statement 2 is incorrect:** The Preamble cannot be directly enforced in a court of law because it is non-justiciable. However, it has been recognised as an aid to the interpretation of the Constitution, useful in resolving ambiguities, and the Supreme Court has held it to be a part of the Constitution and a part of the basic structure.
- The Preamble declares India to be a sovereign, socialist, secular, democratic republic and sets out the objectives of justice, liberty, equality, and fraternity. It was held not to be a part of the Constitution in the *Berubari Union* case but was later held to be a part of it in the *Kesavananda Bharati* case, and it has been amended only once, by the 42nd Amendment Act of 1976.

Q 24. Ans. B

Explanation:

- **Statement 1 is incorrect:** The right to be informed of the grounds of arrest, to consult a legal practitioner, and to be produced before a magistrate within twenty four hours applies to persons arrested under ordinary law; these safeguards do not apply to a person detained under a preventive detention law or to an enemy alien.
- **Statement 2 is correct:** Under Article 22, a person cannot ordinarily be detained under a preventive detention law for more than three months unless an advisory board, consisting of persons qualified to be judges of a High Court, reports sufficient cause for the extended detention before the expiry of three months.
- **Statement 3 is correct:** Preventive detention is a subject on which both Parliament and the state legislatures can legislate, since it appears in the Concurrent List for reasons connected with the security of a state, the maintenance of public order, and the maintenance of supplies and services essential to the community, while Parliament has exclusive power over preventive detention connected with defence, foreign affairs, and the security of India. The 44th Amendment sought to reduce the period of detention without an advisory board to two months, but this provision has not been brought into force.

Q 25. Ans. A

Explanation:

- **Article 38 → Promotion of the welfare of the people by securing a social order**
 - ✓ Article 38 directs the State to promote the welfare of the people by securing and protecting a social order in which justice, social, economic, and political, informs all institutions of national life, and to minimise inequalities in income, status, facilities, and opportunities.
- **Article 39 → Adequate means of livelihood and equal pay for equal work**
 - ✓ Article 39 directs the State to secure adequate means of livelihood for all citizens, the distribution of material resources to subserve the common good, the prevention of concentration of wealth, equal pay for equal work for men and women, and the protection of the health of workers and children.
- **Article 41 → Right to work, education and public assistance**
 - ✓ Article 41 directs the State to secure the right to work, to education, and to public assistance in cases of unemployment, old age, sickness, and disablement, within the limits of its economic capacity.
- **Article 42 → Just and humane conditions of work and maternity relief**
 - ✓ Article 42 directs the State to make provision for just and humane conditions of work and for maternity relief. These belong to the socialist group of directive principles, which also includes Articles 39A, 43, 43A, and 47, while the Gandhian group includes Articles 40, 43B, 46, 47, and 48.

Q 26. Ans. C

Explanation:

- K. C. Wheare described the Indian Constitution as quasi-federal and as a system of government that is federal in form but with a strong unitary bias, because of the predominance of the Centre over the states in legislative, administrative, and financial matters.
- Cooperative federalism does not imply the states functioning independently of the Centre; on the contrary, it implies the Centre and the states working together in a spirit of partnership and collaboration to achieve common national goals,

sharing responsibilities and resources rather than operating in isolation.

- The framers described the system as federal in form but unitary in spirit, and the Constitution itself avoids the word federation, using the expression Union of States. Competitive federalism, by contrast, refers to competition among the states to attract investment and improve governance, while institutions such as the GST Council and bodies promoting joint decision making reflect the cooperative dimension of Indian federalism.

Q 27. Ans. A

Explanation:

- **Statement 1 is correct:** When a vacancy occurs in the office of the President due to death, resignation, removal, or otherwise, the Vice-President acts as President until a new President is elected, and a fresh election must be held within six months, so the Vice-President can act as President only for a maximum period of six months.
- **Statement 2 is incorrect:** Unlike the American Vice-President, who succeeds to the office of President for the remainder of the unexpired term, the Indian Vice-President does not succeed to the office; he merely discharges the functions of the President until a new President is elected. While discharging these functions the Vice-President does not act as the Chairman of the Rajya Sabha and is entitled to the emoluments of the President.
- The Vice-President is elected by an electoral college of the members of both Houses of Parliament, including nominated members, by proportional representation through a single transferable vote. He is the ex officio Chairman of the Rajya Sabha, a position similar to that of the American Vice-President, who is the ex officio Chairman of the Senate.

Q 28. Ans. A

Explanation:

- **Statement 1 is correct:** Article 144 provides that all authorities, civil and judicial, in the territory of India shall act in aid of the Supreme Court, which strengthens the authority of its decisions and ensures their enforcement throughout the country.
- **Statement 2 is correct:** Article 145 empowers the Supreme Court, with the approval of the President, to make rules for regulating its practice and procedure, including rules on the minimum number of judges to sit for deciding cases involving

a substantial question of law as to the interpretation of the Constitution, for which at least five judges must sit.

- **Statement 3 is incorrect:** The Supreme Court can review its own judgments or orders under Article 137, and this power gave rise to the device of the curative petition, by which the Court may reconsider a final decision to prevent abuse of process and a miscarriage of justice. The Court is also a court of record under Article 129 with the power to punish for its contempt.

Q 29. Ans. B

Explanation:

- On a money bill, the Legislative Council enjoys very limited power; it can detain the bill for a maximum period of fourteen days and can only make recommendations, which the Legislative Assembly may accept or reject, so the will of the Assembly ultimately prevails.
- There is no provision for a joint sitting to resolve a deadlock between the two Houses of a state legislature, unlike at the Centre, where Article 108 provides for a joint sitting of the two Houses of Parliament. In a state, the Legislative Council can at most delay an ordinary bill for a few months, after which the view of the Assembly prevails.
- The Legislative Council is thus a subordinate and weaker chamber. The position of the Council as against the Assembly is much weaker than that of the Rajya Sabha as against the Lok Sabha, because the Rajya Sabha enjoys equal powers with the Lok Sabha in many respects, whereas the Council is largely an advisory and dilatory body that the Assembly can override.

Q 30. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 215, a High Court is a court of record, which means that its judgments, proceedings, and acts are recorded for perpetual memory and evidentiary value, and it has the power to punish for its contempt.
- **Statement 2 is correct:** To be appointed a judge of a High Court under Article 217, a person must be a citizen of India and must have held a judicial office in India for ten years, or have been an advocate of a High Court for ten years, so an advocate must have ten years' standing at the bar.
- **Statement 3 is incorrect:** A retired permanent judge of a High Court cannot plead or act in any

court or before any authority in India except the Supreme Court and the other High Courts, so the claim that he can plead in any court is wrong. The independence of the High Court is secured through the mode of appointment, security of tenure, fixed service conditions, salaries charged on the consolidated fund of the state, and a ban on practice after retirement.

Q 31. Ans. A

Explanation:

- **Statement 1 is correct:** India is characterised by a multi-party system, in which a large number of national, state, and regional parties compete for power, a feature attributed to the diversity of the country in terms of religion, region, language, caste, and ideology.
- **Statement 2 is incorrect:** A recognised party, whether national or state, is allotted a reserved symbol, not a free symbol; free symbols are allotted to registered but unrecognised parties and to independent candidates. The reservation of a symbol for a recognised party means that no other candidate can use that symbol.
- On the basis of poll performance, the Election Commission recognises parties as national or state parties, and recognised parties enjoy benefits such as a reserved symbol, free broadcast time on state-owned media, copies of the electoral rolls, and a larger number of star campaigners. Party systems are generally classified into one-party, two-party, and multi-party systems, with India falling in the last category.

Q 32. Ans. A

Explanation:

- **Statement 1 is correct:** Elections to the Lok Sabha and to the state legislative assemblies are held on the first past the post system, under which the candidate who secures the largest number of votes in a constituency is declared elected, even without an absolute majority.
- **Statement 2 is correct:** The Returning Officer is the officer responsible for the conduct of the election in a constituency, including the receipt of nominations, their scrutiny, the conduct of the poll, the counting of votes, and the declaration of the result.
- **Statement 3 is incorrect:** India does not follow proportional representation for the Lok Sabha; proportional representation by a single transferable

vote is used for the election of members of the Rajya Sabha by the assemblies, and for the election of the President and the Vice-President. The election process passes through stages of notification, nomination, scrutiny, withdrawal, campaigning, polling, counting, and declaration of results.

Q 33. Ans. A

Explanation:

- **Statement 1 is correct:** The 100th Constitutional Amendment Act of 2015 gave effect to the acquisition and transfer of certain territories between India and Bangladesh under the land boundary agreement, through the exchange of enclaves and the retention of adverse possessions, and accordingly amended the First Schedule.
- **Statement 2 is incorrect:** The cession of Indian territory to a foreign country cannot be done by a law under Article 3; the Supreme Court held in the Berubari Union case that such a cession requires an amendment of the Constitution under Article 368, since it alters the territory of India in a manner that Article 3 does not contemplate.
- Article 3 empowers Parliament to form new states and to alter the areas, boundaries, or names of existing states by a law passed by a simple majority, on the prior recommendation of the President. The settlement of a boundary dispute, however, does not require a constitutional amendment and can be effected by executive action, a distinction drawn by the Supreme Court.

Q 34. Ans. A

Explanation:

- **Statement 1 is correct:** Unlike Britain, which has a hereditary monarch as the head of the State, India has an elected head of the State in the President, which makes the Indian polity a republic, whereas Britain is a constitutional monarchy.
- **Statement 2 is incorrect:** India does not follow the British principle of unlimited parliamentary sovereignty; the Indian Parliament operates within a written constitution, is subject to judicial review, must respect the fundamental rights, and functions within a federal distribution of powers, so its sovereignty is limited and qualified, unlike the supremacy enjoyed by the British Parliament.
- Other differences are that a British minister must be a member of Parliament, whereas an Indian minister can remain a minister for six months without being a member of either House, and the British system

has the institution of the shadow cabinet, which is not a formal feature of the Indian system. Both systems, however, share the features of a nominal and a real executive, collective responsibility, and the leadership of the Prime Minister.

Q 35. Ans. A

Explanation:

- **Statement 1 is correct:** Citizenship by birth has been made progressively more restrictive; a person born in India between 26 January 1950 and 1 July 1987 is a citizen by birth irrespective of the nationality of the parents, but for those born on or after 3 December 2004, the conditions require that both parents be citizens, or one a citizen and the other not an illegal migrant.
- **Statement 2 is correct:** Loss of citizenship under the Citizenship Act of 1955 can occur in three ways, namely by renunciation, where a citizen voluntarily gives up citizenship, by termination, where a citizen voluntarily acquires the citizenship of another country, and by deprivation, where the government compulsorily terminates citizenship on grounds such as fraud or disloyalty.
- **Statement 3 is incorrect:** A citizen who voluntarily acquires the citizenship of another country does not retain Indian citizenship; the citizenship is automatically terminated, consistent with the principle of single citizenship and with Article 9, which provides that a person who voluntarily acquires foreign citizenship is not a citizen of India. India does not permit dual citizenship, and the Overseas Citizen of India scheme is not a form of dual citizenship.

Q 36. Ans. B

Explanation:

- **21st March:** Known as the Vernal (Spring) Equinox in the Northern Hemisphere.
- **23rd September:** Known as the Autumnal Equinox in the Northern Hemisphere.
- These two dates are known as **Equinoxes** (meaning "equal night" in Latin). During an equinox, the Earth's axis does not tilt toward or away from the Sun, causing the Sun's direct rays to fall exactly on the Equator.

Q 37. Ans. A

Explanation:

- **Statement 1 is correct:** Composite volcanoes, also known as strato-volcanoes, are tall, steep,

cone-shaped mountains. They are built up by explosive eruptions that form alternating layers (strata) of viscous lava, volcanic ash, and cinders. Classic examples include Mount Fuji and Mount Vesuvius.

- **Statement 2 is incorrect:** This statement actually describes **Shield Volcanoes** (like Mauna Loa in Hawaii). Composite volcanoes are formed by highly viscous, silica-rich (andesitic or rhyolitic) lava that does not flow far before cooling, which is why they form steep cones rather than broad, gently sloping landforms.

Q 38. Ans. B

Explanation:

- **Milpa** is the local name for **Shifting Agriculture** (also known as slash-and-burn agriculture) practiced in Mexico and Central America.
- In this ancient farming system, a patch of forest is cleared by felling and burning the trees. The ashes add nutrients to the soil, and crops (like maize, beans, and squash) are grown until the soil loses its fertility, after which the plot is abandoned for a new one.
- **Other regional names for shifting agriculture:** *Jhumming* (North-East India), *Roca* (Brazil), *Ladang* (Indonesia/Malaysia), and *Ray* (Vietnam).

Q 39. Ans. A

Explanation:

- **Statement 1 is correct:** The National Board for Wildlife (NBWL) is a statutory organization constituted under the Wildlife Protection Act of 1972. It is chaired by the **Prime Minister**, making it the apex body for all wildlife-related matters and the approval of projects in and around protected areas.
- **Statement 2 is incorrect:** Originally, forests and the protection of wild animals were in the State List. However, the **42nd Constitutional Amendment Act of 1976** transferred "Forests" and "Protection of Wild Animals and Birds" to the **Concurrent List**, allowing both the Centre and the States to legislate on them.

Q 40. Ans. C

Explanation:

- **Nellore**, a coastal district in Andhra Pradesh, is widely recognised as the "Shrimp Capital of India" (or the Aquaculture Capital of India).
- It earned this title due to its massive, highly productive aquaculture industry, particularly its

extensive brackish water shrimp farming (primarily the cultivation of the Vannamei and Tiger prawn varieties), which contributes heavily to India's seafood exports.

Q 41. Ans. B

Explanation:

- **Statement 1 is correct:** Mixed cropping involves growing two or more crops simultaneously on the same piece of land (e.g., wheat + gram). In rain-fed or dryland areas, combining cereals with drought-resistant pulses is a traditional strategy to minimize the risk of total crop failure due to erratic monsoons.
- **Statement 2 is incorrect:** Mixed cropping is primarily a feature of **subsistence agriculture** aimed at ensuring food security and mitigating weather/pest risks for the farmer's family. In contrast, **commercial agriculture** typically relies on monoculture (growing a single specialised crop over a large area) to maximise yields and streamline harvesting for market sales.

Q 42. Ans. D

Explanation:

- **Salal Project (2):** A run-of-the-river hydroelectric power project on the **Chenab River** in the Reasi district of Jammu & Kashmir.
- **Nagarjuna Sagar Project (4):** One of the world's largest masonry dams, built across the **Krishna River** on the border of Andhra Pradesh and Telangana.
- **Pochampad Project (1):** Also known as the Sri Ram Sagar Project, it is a major irrigation project across the **Godavari River** in Telangana.
- **Shivasamudram Dam (3):** Located in Karnataka on the **Kaveri (Cauvery) River**, site of one of the first hydroelectric power stations in Asia (established in 1902).

Q 43. Ans. B

Explanation:

- A **Glacial Lake Outburst Flood (GLOF)** occurs when the dam containing a glacial lake (usually formed of loose moraine debris or ice) fails, causing a sudden and catastrophic release of massive volumes of water and debris downstream.
- These are highly destructive events and are becoming increasingly common in the Himalayas (e.g., the 2023 South Lhonak Lake disaster in Sikkim) due to rapid glacial retreat caused by climate change.

Q 44. Ans. A**Explanation:**

- The **Green Line** refers to the 1949 Armistice borders established between Israel and its Arab neighbours (Egypt, Jordan, Lebanon, and Syria) following the 1948 Arab-Israeli War.
- It effectively demarcates the pre-1967 borders between the State of Israel and the Palestinian territories (specifically the West Bank and the Gaza Strip). It gets its name from the green ink used to draw the boundary on maps during the armistice talks.

Q 45. Ans. B**Explanation:**

- The **Golden Quadrilateral (GQ)** is India's largest highway project, connecting the four major metropolitan hubs of the country in a quadrilateral shape.
- **Starting from the North and moving in a continuous anti-clockwise sequence, the route connects: Delhi (2) → Mumbai (1) → Chennai (4) → Kolkata (3) before returning to Delhi.**

Q 46. Ans. D**Explanation:**

- **Annular Drainage** is a pattern in which subsequent streams follow a roughly circular or concentric path. This occurs when streams flow along weaker rock outcrops arranged in rings around a central, maturely dissected structural dome or basin.
- In India, this highly specific drainage pattern is classically observed over the domal structures of the **Nilgiri Hills** in Tamil Nadu, parts of the Kerala highlands, and the **Pithoragarh** district of Uttarakhand.

Q 47. Ans. C**Explanation:**

- The **Kumarakom Bird Sanctuary** (also known as the Vembanad Bird Sanctuary) is situated on the eastern banks of **Vembanad Lake** in the Kottayam district of Kerala.
- Vembanad is the longest lake in India and the largest in Kerala. The sanctuary is famous for its backwater ecosystem and is a major wintering ground for migratory birds like the Siberian crane.

Q 48. Ans. A**Explanation:**

- **Statement 1 is correct:** Nilgiri Gneiss is a prominent

geological formation belonging to the **Charnockite rock series** (granulite facies). It is characterised by its bluish-grey to dark appearance and medium to coarse-grained texture, having formed under extremely high temperatures and pressures deep within the earth's crust.

- **Statement 2 is incorrect:** Nilgiri Gneiss was not formed during the Mesozoic Era, nor through volcanic activity. It is one of the oldest rock formations in India, formed during the Archaean (Pre-Cambrian) Era, more than 2.5 billion years ago. It is a metamorphic rock, not a volcanic one, produced by the high-grade metamorphism of pre-existing rocks at great depths in the crust. The Mesozoic Era, by contrast, dates from only about 252 to 66 million years ago, and the major volcanic event of that period in India was the formation of the Deccan Traps (basalts) at the end of the Cretaceous, which is geologically very different from Nilgiri Gneiss.

Q 49. Ans. B**Explanation:**

- In the complex tectonic architecture of the Himalayas, an **autochthonous belt** refers to a block of rocks that remains rooted to its original site of formation, despite surrounding tectonic movement.
- In the Shimla region of the Lesser Himalayas, this narrow belt of highly compressed rocks (ranging from Upper Palaeozoic to Eocene age) is structurally bounded and associated with the **Giri Thrust**. This thrust fault distinctly separates the autochthonous zone from the overriding Krol belt.

Q 50. Ans. B**Explanation:**

- **Pair 1 is incorrect:** The **Drang Drung** glacier is located in the Zaskar Range of Ladakh (Trans-Himalayas), not in the Eastern Himalayas.
- **Pair 2 is incorrect:** The **Shafat** glacier is located in the Suru Valley of the Zaskar Range (Trans-Himalayas), not in the Karakoram Range.
- **Pair 3 is correct:** The **Rimo** glacier is located in the Karakoram Range of Ladakh, forming part of the broader Siachen/Rimo Muztagh glacier system.
- **Pair 4 is correct:** The **Hispar** glacier is located in the Karakoram Range within the Gilgit-Baltistan region.

Q 51. Ans. D

Explanation:

- **Nivedita Dubey → 2 (First Woman Director, Airports Authority of India):** Nivedita Dubey assumed charge as Member (Human Resources) of the Airports Authority of India on 30 January 2026, becoming the first woman to be on the AAI Board in this role.
- **Tarique Rahman → 1 (Prime Minister of Bangladesh):** Tarique Rahman, leader of the Bangladesh Nationalist Party (BNP), was sworn in as Prime Minister of Bangladesh on 17 February 2026 after the BNP's landslide win in the 2026 general election (the first since the 2024 uprising that ended Sheikh Hasina's rule). He is the son of former President Ziaur Rahman and former PM Khaleda Zia.
- **Mia Mottley → 4 (Prime Minister of Barbados, 3rd consecutive term):** Mia Mottley won a historic third consecutive term as Prime Minister of Barbados after her Barbados Labour Party swept all 30 seats in the 11 February 2026 general election.
- **José María Balcázar → 3 (Interim President of Peru):** José María Balcázar, an 83-year-old former judge from the Perú Libre bloc, was elected interim President of Peru by Congress on 18 February 2026, following the removal of José Jerí.

Q 52. Ans. C

Explanation:

- **Statement 1 is correct:** Hope Island is a relatively young, tadpole-shaped sandspit island located off the coast of Kakinada in Andhra Pradesh. It was formed over the last 200 years from sediment deposited by the Godavari River. The island acts as a crucial natural breakwater, protecting the city of Kakinada and Kakinada Bay from the full force of Bay of Bengal cyclones, storm surges, and sea erosion. Because of this natural barrier, Kakinada Port is one of the safest natural ports on India's eastern coast.
- **Statement 2 is correct:** Hope Island is an ecologically sensitive zone and forms a part of the **Coringa Wildlife Sanctuary**, which is situated in the Godavari estuary. The sanctuary boasts the second-largest stretch of mangrove forests in India (after the Sundarbans in West Bengal). The sandy beaches of Hope Island are globally recognised as a critical nesting ground for the vulnerable **Olive Ridley turtles**. Furthermore, its

rich estuarine ecosystem provides a vital feeding and resting ground for numerous migratory birds, including recent notable sightings of Greater Flamingos.

Q 53. Ans. A

Explanation:

- **Statement 1 is correct:** The Network Readiness Index (NRI) is a leading global index that evaluates how well economies leverage information and communication technologies (ICTs) to promote economic growth, innovation, and digital inclusion. It is prepared and released by the **Portulans Institute**, an independent, non-profit research and educational institute based in Washington, D.C. The 2025 report evaluated 127 economies across four main pillars: Technology, People, Governance, and Impact.
- **Statement 2 is incorrect:** While India did indeed improve its position by four ranks, its actual rank in the Network Readiness Index 2025 is **45th** (up from 49th in the 2024 index), not 35th. India's overall score improved from 53.63 to 54.43 out of 100, and it ranked **2nd among lower-middle-income countries**. The country secured the 1st rank globally in specific key indicators such as "Annual investment in telecommunication services", "AI scientific publications", and "ICT services exports".

Q 54. Ans. B

Explanation:

- The autobiography titled '**Diyaslai**' (which translates to 'Matchstick' in Hindi) was authored by the renowned Indian social reformer and 2014 Nobel Peace Prize laureate, **Kailash Satyarthi**. The book was officially launched in early 2025.
- **In this memoir, Satyarthi chronicles his extraordinary journey from a small town in Vidisha, Madhya Pradesh, to becoming a global leader in the fight against child exploitation. The title 'Diyaslai' serves as a powerful metaphor for his philosophy: just as a single small matchstick can pierce through centuries of darkness and ignite a larger fire, a small act of courage and compassion can spark a massive movement for child rights and social justice.** The book details his decades-long struggle, primarily through the *Bachpan Bachao Andolan* (Save the Childhood Movement), which he founded in 1980 to rescue children from forced labour and slavery.

Q 55. Ans. A**Explanation:**

- **Statement 1 is correct:** The Corruption Perceptions Index (CPI) is an annual flagship publication released by **Transparency International**, a prominent global non-governmental organisation dedicated to combating corruption. The 2025 edition of the index evaluated and ranked **182 countries and territories** worldwide based on how corrupt their public sectors are perceived to be by experts and business executives.
- **Statement 2 is incorrect:** The scale operates in the exact opposite manner to what is described in the statement. The CPI uses a scale ranging from **0 to 100**, where **0 indicates highly corrupt** and **100 indicates very clean** (least corrupt). A higher score, therefore, denotes greater transparency and better governance.
- **Statement 3 is correct:** In the CPI 2025, **Denmark** once again topped the global index with a high score of 89, cementing its position as the world's least corrupt country. It was closely followed by Finland (88) and Singapore (84). At the other end of the spectrum, countries experiencing severe conflict, instability, or institutional failure, such as South Sudan (9), Somalia (9), and Venezuela (10), were ranked at the very bottom as the most corrupt.

Q 56. Ans. C**Explanation:**

- **Heavy water (D₂O)** is water in which the hydrogen atoms are replaced by deuterium (²H or D), an isotope of hydrogen. Due to the presence of the heavier isotope, heavy water exhibits physical properties different from those of ordinary water (H₂O).
- The statement "**It has lower density than ordinary water**" is **incorrect** because heavy water is actually **denser than ordinary water**. The density of heavy water is about **1.105 g cm⁻³**, whereas the density of ordinary water is approximately **1.0 g cm⁻³** at room temperature.

Q 57. Ans. B**Explanation:**

- **Rhizopus** is a common fungus that grows on moist bread and is therefore popularly known as **bread mold** or **black bread mold**. It belongs to the group of fungi that reproduce by spores and develops a cotton-like mycelium on the surface of bread, fruits, and other food materials. Rhizopus obtains

its nutrition by decomposing dead and decaying organic matter, making it a saprophytic fungus.

- **Aspergillus** – Commonly found in soil and decaying vegetation; some species are used in industrial production of organic acids.
- **Penicillium** – Known for producing the antibiotic **penicillin** and used in cheese making.
- **Saccharomyces** – Commonly called **yeast**; used in baking and fermentation of alcohol.

Q 58. Ans. D**Explanation:**

- The evolution of plants shows a gradual increase in structural complexity, development of vascular tissues, and adaptation to terrestrial life. **Bryophytes** are the simplest land plants and are often called the "amphibians of the plant kingdom" because they require water for sexual reproduction. They lack true roots, stems, leaves, and vascular tissues.
- **Pteridophytes** evolved after bryophytes and are the first terrestrial plants to possess well-developed vascular tissues (xylem and phloem). They have true roots, stems, and leaves, making them more advanced than bryophytes.
- **Gymnosperms** evolved later and are the first seed-producing plants. Their seeds are "naked" because they are not enclosed within fruits. The development of seeds represents a major evolutionary advancement, reducing dependence on water for reproduction.
- **Angiosperms** are the most advanced and diverse group of plants. They produce flowers and fruits, with seeds enclosed within fruits. Double fertilization and efficient vascular tissues contribute to their evolutionary success and dominance among present-day plants.
- **Therefore, the correct evolutionary sequence from simplest to most complex is:**
 - ✓ **Bryophyte > Pteridophyte > Gymnosperm > Angiosperm.**

Q 59. Ans. D**Explanation:**

- **Statement:** Collenchyma tissues provide mechanical strength to young stems and leaves.
- **Conclusion I:** collenchyma consists of **living cells** that are elongated and irregularly thickened at the corners due to the deposition of cellulose, hemicellulose, and pectin. These unevenly thickened cell walls provide both **mechanical support and flexibility** to young growing parts

such as stems, petioles, and leaves. Therefore, **this conclusion correctly explains how collenchyma provides strength to young plant organs.**

- **Conclusion II: Incorrect**, collenchyma is a **living mechanical tissue**. Unlike sclerenchyma, whose cells are dead at maturity and possess heavily lignified walls, collenchyma cells remain alive even after maturity. Hence, this conclusion is not correct.

Q 60. Ans. C

Explanation:

- Chlamydomonas → Green algae
 - ✓ **Chlamydomonas** is a unicellular, freshwater alga belonging to the class **Chlorophyceae (Green Algae)**. It contains chlorophyll a and b, giving it a characteristic green color.
- Laminaria → Brown algae
 - ✓ **Laminaria** (kelp) belongs to the class **Phaeophyceae (Brown Algae)**. Laminaria is an example of brown algae, characterized by the pigment **fucoxanthin**, which gives them a brown coloration.
- Gelidium → Red algae
 - ✓ **Gelidium** belongs to the class **Rhodophyceae (Red Algae)**. Gelidium is an important red alga used commercially for the extraction of **agar**.

Q 61. Ans. C

Explanation:

- **Binary fission** is an asexual mode of reproduction in which a single parent cell divides into two nearly equal daughter cells. **Amoeba** reproduces by binary fission under favorable conditions.
- **Budding** is a form of asexual reproduction in which a small outgrowth or bud develops on the parent body and eventually separates. **Yeast** as a classic example of reproduction by budding.
- **Fragmentation** occurs when the body of an organism breaks into fragments, and each fragment develops into a new individual. The filamentous green alga **Spirogyra** reproduces by fragmentation.
- **Spore formation** is a common asexual reproductive method in fungi. fungi such as **Penicillium** produce spores that can germinate under favorable conditions to form new individuals.

Q 62. Ans. B

Explanation:

- **Statement:** Anaerobic respiration produces less ATP compared to aerobic respiration.
- **Conclusion I:** anaerobic respiration involves the

incomplete breakdown of glucose in the absence of oxygen. The end products are substances such as **ethanol and carbon dioxide (in yeast) or lactic acid (in muscle cells)**. Since glucose is **not completely oxidized to carbon dioxide** and water, much of the energy remains stored in the end products. **Therefore, this conclusion is incorrect.**

- **Conclusion II:** during anaerobic respiration, ATP is produced only through **glycolysis**, which yields a net gain of **2 ATP molecules per glucose molecule**. In contrast, aerobic respiration completely oxidizes glucose and produces a much larger amount of ATP. This low ATP yield is the reason anaerobic respiration is less efficient than aerobic respiration.

Q 63. Ans. D

Explanation:

- Sieve tube elements are highly specialised, elongated cells found in the phloem (the vascular tissue responsible for transporting food and sugars) of flowering plants.
- During their maturation process, these cells undergo a selective internal breakdown, losing their nucleus, ribosomes, and large vacuoles. This evolutionary adaptation creates a wide-open, unobstructed channel with low resistance, allowing nutrient-rich sap to flow rapidly and efficiently.
- Because they lack a nucleus and ribosomes, mature sieve tube elements cannot synthesise proteins, produce energy independently, or manage their own cellular maintenance. They survive only because of their strict, symbiotic dependency on adjacent **companion cells**.
- A sieve tube element and its adjacent companion cell are derived from the exact same mother cell during division.
- The companion cell retains a large, active nucleus and abundant organelles. Its nucleus effectively acts as the "brain" for both cells, controlling the metabolic activities required to keep the enucleated (nucleus-free) sieve tube element alive.
- The two cells are intimately connected by numerous plasmodesmata (microscopic channels that cross their shared cell walls). The companion cell continuously pumps essential proteins, ATP (cellular energy), and signalling molecules directly into the sieve tube element through these channels.

Q 64. Ans. B

Explanation:

- Lysosomes are small, membrane-bound organelles

found in animal cells (and some plant cells) that act as the cell's waste disposal system. They are filled with highly potent **hydrolytic (digestive) enzymes** capable of breaking down all types of biological macromolecules, including proteins, lipids, carbohydrates, and nucleic acids.

- Under normal conditions, lysosomes use these enzymes to digest foreign materials (like bacteria or viruses) and recycle worn-out cell organelles, a process known as autophagy.
- However, they earn the nickname "**suicidal bags**" (or "suicide sacs") because of what happens during severe cellular damage, disease, or cellular ageing. If a cell is damaged beyond repair, the membrane of the lysosome can rupture. When this happens, the powerful digestive enzymes spill out into the cytoplasm and digest the cell's own components, ultimately destroying the entire cell. This programmed self-destruction process is called **autolysis**.

Q 65. Ans. B

Explanation:

- **Petal → Attract insects**
 - ✓ Petals are usually brightly coloured and often fragrant structures of a flower. Their primary function is to attract insects, birds, and other pollinators that help in the process of pollination. The colourful petals make flowers conspicuous and facilitate the transfer of pollen from one flower to another. **Therefore, Petal is correctly matched with "Attract insects."**
- **Stamen → Produces pollen**
 - ✓ The stamen is the male reproductive part of a flower. It consists of a filament and an anther. The anther contains pollen grains, which carry the male gametes required for reproduction. pollen grains are produced in the anther and are transferred to the stigma during pollination. Hence, **Stamen is correctly matched with "Produces pollen."**
- **Sepal → Protects flower**
 - ✓ Sepals are the outermost green leaf-like structures of a flower that collectively form the calyx. Their main role is to protect the flower when it is in the bud stage. They shield the developing reproductive organs from physical damage and adverse environmental conditions. **Therefore, Sepal is correctly matched with "Protects flower."**
- **Pistil → Female reproductive**

- ✓ The pistil, also known as the carpel, is the female reproductive organ of a flower. It consists of the stigma, style, and ovary. The ovary contains ovules, which develop into seeds after fertilization. The stigma receives pollen grains during pollination. **Thus, Pistil is correctly matched with "Female reproductive part."**

Q 66. Ans. B

Explanation:

- **The Rahnumai Mazdayasnan Sabha was the most important socio-religious reform movement among the Parsis in the 19th century. It was founded in 1851 in Bombay by reformers such as:**
 - ✓ Naoroji Furdonji
 - ✓ Dadabhai Naoroji
 - ✓ K.R. Cama
- **The Sabha aimed to:**
 - ✓ Purify Zoroastrian religion.
 - ✓ Promote modern education.
 - ✓ Improve the status of women.
 - ✓ Encourage social reforms such as female education and opposition to outdated customs.
- To spread these reformist ideas, Dadabhai Naoroji started the journal *Rast Goftar* ("Truth Teller"), which became the principal organ of the movement and the chief medium for disseminating Parsi socio-religious reform ideas.

Q 67. Ans. C

Explanation:

- **Statement 1 is correct:** Cornwallis sought to establish a clearer separation between revenue collection and the administration of justice. The District Collector was largely deprived of judicial and magisterial functions, and separate judicial officers were appointed to administer civil and criminal justice.
- **Statement 2 is incorrect:** The Sadar Diwani Adalat (highest civil court) and Sadar Nizamat Adalat (highest criminal court) were not abolished by the Cornwallis Code. In fact, they continued to function as the apex courts within the Company's judicial system. Appeals to the British Privy Council were possible only in certain cases, but the Sadar courts remained intact.
- **Statement 3 is correct:** The Code created a more systematic judicial hierarchy:
 - ✓ Munsif Courts at the lowest level for petty civil suits,

- ✓ Registrar's Courts,
- ✓ District Diwani Adalats,
- ✓ Provincial Courts of Appeal,
- ✓ Sadar Diwani Adalat at the apex of the civil judicial structure.

Q 68. Ans. D

Explanation:

- **Battle of Porto Novo (1781):** During the Second Anglo-Mysore War, the British commander Sir Eyre Coote decisively defeated Haidar Ali at the Battle of Porto Novo on July 1, 1781. This victory was crucial for the British in checking Mysore's advance.
- **Third Anglo-Mysore War (1790–1792):** This war concluded with the defeat of Tipu Sultan by the allied forces of the British (led by Lord Cornwallis), the Marathas, and the Nizam. It ended with the humiliating **Treaty of Seringapatam (1792)**, which forced Tipu Sultan to cede nearly half of his territories to the victors and pay a massive war indemnity.
- **Fourth Anglo-Mysore War (1799):** Fought under Governor-General Lord Wellesley, this brief and decisive war resulted in the death of Tipu Sultan while defending his capital, Seringapatam. It completely destroyed Mysore's independence; the British annexed large portions of the state and restored the core region to the previous Hindu ruling family (the Wodeyars) under a Subsidiary Alliance.

Q 69. Ans. C

Explanation:

- **Statement 1 is correct:** The regional states that emerged after the decline of the Mughal Empire (such as Awadh, Bengal, Hyderabad, and the Maratha kingdom) were essentially successor states. While they may have reorganized administration or increased the use of revenue farming (ijaradari), none of them fundamentally altered the basic rent-extracting, agrarian-based economic structure established by the Mughals. They continued to rely heavily on squeezing surplus from the peasantry.
- **Statement 2 is correct:** Throughout the 18th century (prior to the full impact of the British Industrial Revolution), India was the world's largest manufacturer of cotton textiles. Indian textiles, such as calicoes, muslins, and chintzes were in massive demand across Europe, Southeast Asia, and the Middle East, making them the absolute most important export item from Indian ports.

- **Statement 3 is incorrect:** It was the exact opposite. During this period, India ran a massive trade **surplus**, not a deficit. Because European nations had very few goods that Indians actually wanted to buy, the European East India Companies had to pay for Indian textiles and spices using precious metals. Consequently, India was historically known as the "sink" of the world's bullion, importing massive amounts of gold and silver rather than exporting it.

Q 70. Ans. B

Explanation:

- **Assertion (A) is true:** Alivardi Khan was determined to maintain absolute sovereignty over Bengal. He strictly forbade the British, French, and Dutch from building fortifications or mounting guns on their factories.
- **Reason (R) is true:** The rampant misuse of dastaks (trade permits meant only for the Company's export/import goods) for the private inland trade of Company officials was a massive, ongoing economic grievance during Alivardi Khan's reign.
- **Why (R) does not explain (A):** Alivardi Khan banned fortifications as a "punishment" for the Company's tax evasion. However, his ban on fortifications was an overarching **geopolitical and military policy**, not an economic retaliation. He had keenly observed the Carnatic Wars in South India and realised that if he allowed any Europeans (British or French) to militarise their settlements, they would eventually fight proxy wars on his soil and threaten his throne. He famously compared the Europeans to bees: "If you leave them alone, they will give you honey; but if you disturb them, they will sting you to death." Therefore, the economic friction (R) was not the cause of the military ban (A).

Q 71. Ans. B

Explanation:

- The Rowlatt Act (officially the Anarchical and Revolutionary Crimes Act of 1919) was deeply resented precisely because it bypassed foundational principles of British justice.
- The Act directly suspended the right of **Habeas Corpus**, a bedrock legal principle of English common law that protects individuals against unlawful and indefinite imprisonment by requiring that anyone arrested be brought before a judge or court.
- The Act authorised the British colonial government to arrest Indians suspected of treason or anti-colonial activities and detain them indefinitely.

without trial, without a warrant, and without the right to judicial review for up to two years.

- **The suspension of basic legal rights led to massive nationwide protests. The sheer denial of legal representation and judicial process was famously summarized by the popular Indian slogan of the time: "No Dalil, No Vakil, No Appeal" (No arguments, no lawyer, no appeal).**

Q 72. Ans. C

Explanation:

- **Young India** was an English weekly journal published by Mahatma Gandhi from 1919 to 1931. It served as his primary platform to spread the philosophy of non-violence (Ahimsa) and Satyagraha to the English-reading Indian masses and the British authorities.
- In 1920, against the backdrop of the Jallianwala Bagh massacre and the Khilafat issue, Gandhi wrote extensively in *Young India* calling for Non-Cooperation. It was in this journal (in his famous August 1920 article, "The Doctrine of the Sword") that he defended India's honour, famously writing that he would rather have India resort to arms to defend her honour than cowardly become a helpless witness to her own dishonour.
- **Indian Opinion (1903):** Founded by Gandhi in South Africa to fight for the civil rights of the Indian diaspora.
- **Navajivan (1919):** A weekly journal similar to *Young India*, but published in Gujarati.
- **Harijan (1933):** Launched much later by Gandhi to campaign against the social evil of untouchability.

Q 73. Ans. A

Explanation:

- Mustafa Kamal Pasha (later known as **Mustafa Kemal Atatürk**) was a visionary Turkish nationalist, military commander, and the founding father of the Republic of Turkey.
- Following the defeat of the Ottoman Empire in World War I, the victorious Allied powers sought to partition Turkey. Mustafa Kamal led a successful War of Independence against them. In November 1922, his nationalist government in Ankara abolished the centuries-old Ottoman Sultanate, stripping Sultan Mehmed VI of his political power.
- Kamal Pasha transformed Turkey into a modern, secular nation-state. He went on to abolish the Caliphate entirely in 1924 (which ironically led to the collapse of the Khilafat Movement in India). His

sweeping reforms included adopting Western legal codes, replacing the Arabic script with the Latin alphabet, closing religious courts, and granting equal political rights to women.

Q 74. Ans. D

Explanation:

- On February 5, 1922, a violent mob set fire to a police station at Chauri Chaura (in Gorakhpur, UP), burning 22 policemen alive. Deeply disturbed by this breach of non-violence, Mahatma Gandhi immediately decided to halt the movement.
- **Bardoli Resolution:** On February 12, 1922, the Congress Working Committee (CWC) convened at **Bardoli**, Gujarat. Here, they formally passed the "Bardoli Resolution," which immediately suspended all active, law-breaking campaigns (civil disobedience and strikes) across the country.
- To channel the energy of the masses, the resolution instructed Congress workers to stop agitating and instead dedicate themselves to a peaceful "Constructive Programme." This included promoting khadi (hand-spinning), national education, Hindu-Muslim unity, and the eradication of untouchability.
- **Calcutta** (Sept 1920) and **Nagpur** (Dec 1920) were the crucial Congress sessions where the Non-Cooperation Movement was initially proposed and finally ratified.

Q 75. Ans. A

Explanation:

- **Statement 1 is incorrect:** The Nehru Report (1928) was drafted by an all-parties committee chaired by **Motilal Nehru** (the father), not Jawaharlal Nehru. While Jawaharlal Nehru acted as the secretary of the committee, his father, Motilal, was its chief architect. (In fact, Jawaharlal Nehru and Subhas Chandra Bose actually opposed the report's core demand of "Dominion Status," as they wanted "Complete Independence" or *Purna Swaraj*).
- **Statement 2 is correct:** The Nehru Report was presented at the All-Party Convention held in Calcutta in December 1928. It failed to pass because of fierce disagreements among communal-minded leaders. M.A. Jinnah (representing the Muslim League) demanded amendments regarding Muslim representation and residuary powers. M.R. Jayakar (representing the Hindu Mahasabha) vehemently opposed Jinnah's demands. Due to this communal deadlock, the convention failed to adopt the report.

Q 76. Ans. B**Explanation:**

- The annual session of the Indian National Congress in 1927 was held in Madras and was presided over by **Dr. Mukhtar Ahmed Ansari (Dr. M.A. Ansari)**.
- In November 1927, the British government announced the appointment of the Indian Statutory Commission (the Simon Commission) to review constitutional reforms. Because all seven members of the commission were British (making it an "all-white" commission), Indians felt deeply insulted that their political future was being decided without any Indian representation. Under Dr. Ansari's presidency, the Congress formally resolved to boycott the commission "at every stage and in every form."
- The 1927 Madras session is also historically significant because it was here that a young Jawaharlal Nehru successfully moved a resolution declaring 'Complete Independence' as the ultimate goal of the Congress, setting the stage for the famous Poorna Swaraj resolution two years later in Lahore.

Q 77. Ans. D**Explanation:**

- **Assertion (A) is false:** During the Lahore session in December 1929, the Congress did not draw up a detailed, step-by-step program of struggle. While the session authorised the All India Congress Committee to launch a program of civil disobedience (including the non-payment of taxes), it deliberately left the exact timing, method, and detailed strategy entirely in the hands of Mahatma Gandhi. It was Gandhi who later conceptualized the highly specific Salt Satyagraha as the opening move.
- **Reason (R) is true:** The Lahore session is historically famous precisely because it was here, under the presidency of a young Jawaharlal Nehru, that the Congress officially dropped the demand for "Dominion Status" and passed the historic resolution declaring **Poorna Swaraj** (Complete Independence) as its ultimate and only objective.

Q 78. Ans. B**Explanation:**

- Government of India Act, 1935 proposed an "All-India Federation" consisting of British Indian provinces and Princely States. However, this federal structure could only come into existence if a specified number of Princely States agreed

to join by signing an 'Instrument of Accession'. Since the princes refused to join, fearing a loss of their autocratic powers, the Federal part of the Act was never implemented. The central government continued to operate under the older Government of India Act of 1919.

- The Act successfully introduced "Provincial Autonomy," abolishing the unpopular system of dyarchy in the provinces. This part of the Act was brought into operation, leading to the historic provincial elections of 1937, where the Indian National Congress formed ministries in the majority of the provinces.

Q 79. Ans. C**Explanation:**

- **Statement 1 is correct:** Jawaharlal Nehru presided over the Congress during three highly significant sessions in this era:
 - ✓ **1929 (Lahore):** Famous for the Poorna Swaraj (Complete Independence) resolution.
 - ✓ **1936 (Lucknow):** Where Nehru pushed for socialist policies and agrarian reforms.
 - ✓ **1937 (Faizpur):** Historically significant as the first-ever Congress session held in a rural village, focusing heavily on peasant issues.
- **Statement 2 is correct:** Subhas Chandra Bose was elected President for two consecutive years:
 - ✓ **1938 (Haripura):** Where he set up the National Planning Committee (under Nehru's chairmanship).
 - ✓ **1939 (Tripuri):** Bose was re-elected by defeating Dr. Pattabhi Sitaramayya (who was openly backed by Mahatma Gandhi). This led to the famous "Tripuri Crisis," resulting in Bose resigning shortly after and forming the Forward Bloc.

Q 80. Ans. D**Explanation:**

- The National Planning Committee (NPC) was set up in **1938** during the Haripura session of the Indian National Congress. It was initiated by the Congress President at the time, **Subhas Chandra Bose**, who strongly believed in forced industrialisation and centralised state planning.
- Although Bose initiated the committee, he appointed **Jawaharlal Nehru** as its Chairman. Nehru's deep interest in Soviet-style economic planning and socialist development made him the natural choice to lead this body.

- The committee's goal was to draw up a comprehensive economic plan for a free India, focusing on heavy industries, poverty eradication, and agricultural reform. (Its work was unfortunately interrupted by the outbreak of World War II and the subsequent Quit India Movement).
- Acharya Narendra Dev and Jai Prakash Narayan were founding pillars of the Congress Socialist Party (CSP), while P.C. Joshi was the first general secretary of the Communist Party of India (CPI). None of them chaired the NPC.

Q 81. Ans. D

Explanation:

- For a long time, the Indian National Congress maintained a policy of strict "non-interference" in the internal political affairs of the Princely States. Even as late as the 1938 Haripura session, the Congress resolved that while it supported the states' people, the actual struggles should be conducted by independent local organisations (like the Praja Mandals) rather than in the name of the Congress.
- By 1939, massive mass awakenings and agitations had erupted in several states (like Rajkot, Jaipur, Kashmir, Hyderabad, and Travancore). Recognising this shift, the Congress at the **Tripuri Session (1939)** completely revised its policy. It passed a resolution declaring that the "great awakening" meant the Haripura restrictions were no longer necessary, formally deciding to take an active, direct, and guiding part in the political movements of the States' people.
- This effectively merged the democratic struggles against the autocratic Princes with the national struggle against British imperialism, treating India as one indivisible political unit.

Q 82. Ans. B

Explanation:

- In the Tamluk subdivision of the Midnapore district, revolutionaries successfully established the **Jatiya Sarkar** (National Government), which functioned from December 1942 to September 1944. It was highly organised, undertaking cyclone and famine relief work, setting up schools, and organizing an armed wing called the Vidyut Vahini.
- **Other Parallel Governments:**
 - ✓ **Ballia (Uttar Pradesh):** The first to be established, led by Chittu Pandey, though it only lasted for a week.
 - ✓ **Satara (Maharashtra):** Named the Prati Sarkar,

it was the longest-lasting parallel government of the movement, functioning from mid-1943 to 1945 under leaders like Nana Patil.

Q 83. Ans. D

Explanation:

- The text 'Tuhfa-i-Akbar Shahi' was authored by **Abbas Khan Sarwani**, a historian and noble who was connected to Sher Shah's family by marriage.
- Ironically, this glowing account of the Afghan ruler Sher Shah Suri was actually commissioned by the Mughal Emperor Akbar. Akbar wanted a detailed historical record of Sher Shah's reign, particularly his highly efficient land revenue and administrative systems, which Akbar's own administration (under Todar Mal) subsequently adapted and improved.
- **Abul Fazl** was Akbar's chief historian and wrote the Akbarnama.
- **Faizi** was Akbar's poet laureate and translated mathematical/Sanskrit texts (like the Lilavati).
- **Ralph Fitch** was an English merchant who visited India during Akbar's reign.

Q 84. Ans. B

Explanation:

- In Akbar's administration, the **Diwan-i-Aala** (also referred to simply as the Diwan or Wazir) was the head of the revenue department. He acted as the principal financial officer of the empire. His duties included managing the imperial treasury, auditing accounts, supervising the allocation of *jagirs* (land grants), and overseeing the collection of land revenue across the empire. Famous figures like Todar Mal held this crucial portfolio.
- **Sipahsalar** (later known as the Subahdar) was the governor and military commander of a province (*Subah*).
- **Bakhshi** (specifically the *Mir Bakhshi*) was the head of the military department and the paymaster general of the empire, responsible for the Mansabdari system and intelligence gathering.
- **Faujdar** was the executive head and military commander of a district (*Sarkar*), primarily responsible for maintaining law and order.

Q 85. Ans. A

Explanation:

- **Abdul Ghaffoor Bohra** (often referred to in historical texts as Mulla Abdul Ghafur) was the wealthiest and most influential merchant-prince and shipowner in Surat during the late 17th and

early 18th centuries (primarily during Aurangzeb's reign). He practically monopolised the Indian Ocean trade, operating a massive private fleet of over 20 trading vessels that rivaled the European East India Companies.

- Upon his death in 1718, his estate was famously valued at a staggering 85 lakh rupees in cash and goods, an astronomical sum for a private citizen at the time. (This massive fortune later became a target for the local Mughal governor, who attempted to confiscate a large portion of it from his grandson).
- **Virji Vohra:** He was the richest merchant of Surat a generation earlier (during the reigns of Jahangir and Shah Jahan) and often acted as the primary financier for the British East India Company, but he died around 1670.
- **Malaya Chetti:** A prominent 17th-century merchant and broker who operated on the Coromandel Coast (South India), not the Gujarat/Surat trade hub.
- **Raja Todar Mal:** The famous Finance Minister (Diwan-i-Aala) of Emperor Akbar. He was a statesman and administrator, not a private maritime merchant, and he died in 1589.

Q 86. Ans. A

Explanation:

- According to the data released by the Ministry of Commerce and Industry for April 2026, **coal** production recorded the steepest year-on-year contraction among all eight sectors, falling by **8.7%**. This was primarily due to a high base effect from the previous year and logistical constraints.
- **Fertilizers:** Closely followed coal with the second-largest decline, dropping by 8.6%.
- **Natural Gas & Crude Oil:** These sectors continued their prolonged downward trends, contracting by 4.3% and 3.9% respectively in April 2026.
- **Overall:** The combined Index of Eight Core Industries registered a very modest provisional growth of 1.7% in April 2026. Out of the eight sectors, five contracted, while only three registered positive growth (Cement at 9.4%, Steel at 6.2%, and Electricity at 4.1%).

Q 87. Ans. C

Explanation:

- According to the global **Trade Rankings of Major Countries/Areas in the World, 2025** (compiled from World Trade Organization data), India ranked **13th** overall in total merchandise trade, with a combined trade value of roughly US\$ 1.2 trillion.

- India's 13th-place finish in total trade is a combined result of its heavy import reliance and growing exports. Specifically, in the 2025 rankings, India was the world's **9th** largest importer of merchandise, but the **19th** largest exporter.
- The top three spots for total merchandise trade in 2025 were retained by China (1st), the United States (2nd), and Germany (3rd).
- India performs significantly better in the "Commercial Services" trade rankings compared to physical merchandise.

Q 88. Ans. A

Explanation:

- **Statement 1 is correct:** An absolute poverty line defines a minimum threshold of income needed to secure the basic necessities of life (like food, shelter, and clothing). The World Bank's international poverty line (currently set at \$2.15 a day using 2017 Purchasing Power Parity) is the classic global example of an absolute poverty metric. It represents a fixed standard of survival across different countries.
- **Statement 2 is incorrect:** An absolute poverty line does not automatically rise with overall income growth in the economy. It only adjusts for inflation to maintain the same purchasing power over time. The concept described in Statement 2 is a **relative poverty line** (commonly used in developed European nations), which is typically defined as a percentage of the median national income (e.g., earning less than 50% of the median wage). Because a relative line is tied to the median, it automatically rises as the overall economy grows.

Q 89. Ans. A

Explanation:

- **Statement 1 is incorrect:** While the 'Navratna' status grants CPSEs enhanced financial and operational autonomy (such as the ability to invest up to ₹1,000 crore or 15% of their net worth in a single project without prior government approval), it does not grant "complete autonomy without any government oversight." They remain government-owned entities and are strictly subject to parliamentary oversight, audits by the Comptroller and Auditor General (CAG), and broader guidelines from their respective administrative ministries.
- **Statement 2 is correct:** The 'Navratna' status is strictly awarded based on quantitative performance and profitability. A CPSE must first be a Miniratna

Category-I, have four independent directors on its board, and achieve a composite score of 60 or above out of 100 across six performance parameters (such as net profit to net worth, manpower cost to total cost of production, and earnings per share).

- **Statement 3 is incorrect:** The number of Navratna companies is nowhere near 100. As of 2026, following recent additions (like IRFC, IRCTC, NHPC, and SECI), there are currently **26 CPSEs** holding the 'Navratna' status.

Q 90. Ans. B

Explanation:

- The Worker Population Ratio (WPR) is defined as the percentage of employed persons in the total population. It is a critical indicator of the economy's ability to generate active employment.
- According to the PLFS Annual Report for the calendar year 2025 (released by the National Statistical Office/MoSPI), the WPR for persons aged 15 years and above under the usual status (principal + subsidiary status) was estimated at **57.4%**.
- Out of this 57.4%, the male WPR stood at a robust 76.6%, while the female WPR was recorded at 38.8% (continuing a positive trend of increasing female workforce participation over the last few years).
- For the same period (2025), the overall Labour Force Participation Rate (LFPR) for ages 15 and above stood at 59.3%, and the overall Unemployment Rate (UR) declined to 3.1%.

Q 91. Ans. B

Explanation:

- **Statement 1 is correct:** The GII is explicitly designed to measure gender disparities and the loss of potential human development across three distinct dimensions: **reproductive health, empowerment, and the labour market** (economic status).
- **Statement 2 is correct:** The reproductive health dimension is measured using exactly two indicators: the **maternal mortality ratio (MMR)** and the **adolescent birth rate (ABR)**, also known as the adolescent fertility rate).
- **Statement 3 is incorrect:** While the share of parliamentary seats is indeed an empowerment indicator, it is **not used for women only**. Because the GII measures inequality, it calculates the ratio and disparity between the share of parliamentary seats held by **both women and men**. (The empowerment dimension also includes a second indicator: the proportion of both adult women and men who have

reached at least a secondary level of education).

- **Statement 4 is correct:** To measure the labour market dimension, the index includes the **Labour Force Participation Rate (LFPR) for both females and males aged 15 and older**.

Q 92. Ans. B

Explanation:

- National poverty headcount ratios cannot be calculated by looking at macroeconomic averages; they require granular, household-level data on what people are actually consuming or earning. Since it is impossible to survey every single household in a country every year for detailed expenditure, governments rely on large-scale **household sample surveys** (such as the Household Consumer Expenditure Survey conducted by the NSSO in India). The data from this sample is then multiplied by population weights to estimate the poverty ratio for the entire national population.
- While **population censuses** provide a complete demographic headcount, they do not collect the highly detailed consumption, expenditure, or income data required to calculate poverty lines. Censuses are simply too massive and expensive to include 100-question expenditure modules.
- **National income accounts** (which give us GDP and Per Capita Income) only provide a macroeconomic average. An average per capita income tells nothing about the distribution of that wealth (e.g., whether 10 people have \$100 or 1 person has \$1000). Therefore, it cannot calculate a poverty headcount from it.
- **Labour force surveys** (like India's PLFS) are designed to track employment, unemployment, and workforce participation, not the detailed household consumption baskets needed to determine who falls below a designated poverty line.

Q 93. Ans. B

Explanation:

- **Conditionality:** When the International Monetary Fund (IMF) lends money to a country facing a balance of payments crisis, it doesn't just hand over the cash. It attaches strict conditions, requiring the borrowing government to implement specific macroeconomic and structural policies (such as reducing fiscal deficits, eliminating subsidies, curbing inflation, or privatising state-owned enterprises). The goal is to ensure the country fixes its underlying economic problems and can repay

the loan. This practice is universally known as IMF "conditionality."

- **Quantitative Easing (QE):** An unconventional domestic monetary policy where a central bank purchases large-scale government bonds or other assets to inject liquidity into its own economy.
- **Sterilization:** A central bank operation designed to neutralise the impact of foreign exchange interventions on the domestic money supply.
- **Dollarization:** When a country officially or unofficially adopts a foreign currency (often the US Dollar) for domestic transactions, abandoning or pegging its own currency.

Q 94. Ans. C

Explanation:

- **Statement 1 is correct:** The Most-Favoured-Nation (MFN) principle (enshrined in Article I of the General Agreement on Tariffs and Trade, or GATT) is the cornerstone of the WTO. It essentially means non-discrimination. If a WTO member grants a special favor (such as a lower customs duty rate for one of their products) to any other country, it must immediately and unconditionally grant that same treatment to all other WTO members.
- **Statement 2 is correct:** The philosophical and practical goal of the MFN principle is precisely to prevent discriminatory, power-based bilateral deals where large economies bully smaller ones into exclusive agreements. By requiring that concessions be extended multilaterally to all members, it creates a fairer, rules-based global trading system where smaller economies benefit from the same trade terms negotiated by larger economic powers.

Q 95. Ans. C

Explanation:

- **Statement 1 is correct:** Payments Banks are specifically designed to leverage technology to provide fast, low-cost remittance services, fund transfers, and digital banking facilities to migrant labour workforce, low-income households, and small businesses.
- **Statement 2 is incorrect:** The framework for Payments Banks was conceptualized based on the recommendations of the **Nachiket Mor Committee** (Committee on Comprehensive Financial Services for Small Businesses and Low Income Households). The **Usha Thorat Committee** is associated with the licensing of *Small Finance Banks*, not Payments Banks.

- **Statement 3 is correct:** Payments Banks are structured to hold money but not risk it. They can accept demand deposits (currently capped at ₹2 lakh per individual customer) and issue ATM/debit cards. However, they are strictly prohibited from undertaking lending operations and cannot issue credit cards.

Committee	Core Area / Major Recommendation
Narasimham (1991 & 1998)	Comprehensive Banking Reforms (lowering SLR/CRR, Capital Adequacy, setting up ARCs).
Nachiket Mor (2013)	Creation of Payments Banks for financial inclusion.
Usha Thorat (2014)	Licensing framework for Small Finance Banks (SFBs).
B. Sivaraman (1979)	Establishment of NABARD for rural and agricultural credit.
Urjit Patel (2014)	Revision of Monetary Policy (adoption of Inflation Targeting and the MPC).
P.J. Nayak (2014)	PSB Governance (reducing government stake, creating the Banks Board Bureau).
N.K. Singh (2016)	Review of the FRBM Act (setting strict fiscal deficit and debt-to-GDP targets).
Vijay Kelkar (2002 & 2015)	Tax Reforms and revitalizing the PPP Model for infrastructure projects.
S.S. Tarapore (1997 & 2006)	Roadmap for Full Capital Account Convertibility of the Rupee.
Bimal Jalan (2019)	Economic Capital Framework (deciding RBI's surplus transfer to the government).
Y.H. Malegam (2011)	Framework and regulation for Microfinance Institutions (MFIs).

Q 96. Ans. B

Explanation:

- **Statement 1 is incorrect:** Fungi are **non-green, heterotrophic organisms** that lack chlorophyll. Since chlorophyll is essential for photosynthesis, fungi cannot prepare their own food. Instead, they obtain nutrients from dead and decaying organic matter (saprophytic nutrition), living hosts (parasitic nutrition), or through symbiotic associations. fungi belong to a separate kingdom because they differ fundamentally from plants in their mode of nutrition and absence of chlorophyll.
- **Statement 2 is correct:** Unlike plants, which store food mainly as starch, fungi store reserve food

materials in the form of **glycogen and oils**. This characteristic is one of the important features used to distinguish fungi from plants. Fungi are heterotrophic organisms with cell walls made of chitin and store food as glycogen.

Q 97. Ans. C

Explanation:

- The dark reaction, also known as the **Calvin cycle** or **biosynthetic phase of photosynthesis**, occurs in the **stroma** of the chloroplast. The stroma is the fluid-filled matrix surrounding the grana. During this phase, carbon dioxide is fixed into carbohydrates using the ATP and NADPH produced during the light reaction. Although called the "dark reaction," it does not necessarily occur in darkness; it simply does not require direct light.

Q 98. Ans. C

Explanation:

- The **plasma membrane**, also known as the **cell membrane**, is a thin, living, flexible boundary that surrounds the cell. It is **selectively permeable (semi-permeable)**, meaning it allows only certain substances to enter or leave the cell while restricting others. This property helps maintain the internal environment of the cell and regulates the exchange of nutrients, gases, water, and waste products between the cell and its surroundings.

Q 99. Ans. A

Explanation:

- **Statement 1 is correct:** The Law of Segregation is one of the fundamental laws of inheritance proposed by Gregor Mendel. According to this law, the two alleles of a gene present in an individual separate from each other during gamete formation, so that each gamete receives only one allele. Since the alleles remain distinct and do not blend with each other, the law is also known as the **Law of Purity of Gametes**. This ensures that gametes carry only one factor (allele) for a particular trait.
- **Statement 2 is incorrect:** The Law of Segregation states exactly the opposite. During gamete formation, the pair of alleles separates, and **only one allele enters each gamete**. Thus, a gamete is pure for a particular character. For example, a heterozygous plant with genotype Tt produces two types of gametes, T and t, not gametes containing both T and t together.

Q 100. Ans. B

Explanation:

- The inner lining of the small intestine contains numerous tiny finger-like projections called **villi**. These villi greatly increase the surface area available for the absorption of digested food. Each villus contains a network of blood capillaries and a lymph vessel (lacteal), which help in the efficient absorption and transport of nutrients throughout the body. The large surface area provided by villi ensures maximum absorption of food products.
- **Cilia are microscopic hair-like structures present on the surface of certain cells**. They help in movement or in moving substances over the cell surface. For example, cilia line the respiratory tract and help remove dust and mucus. They are not the finger-like projections found in the intestine.
- **Papillae are small projections found on the tongue and certain other tissues**. Some papillae contain taste buds and help in the sensation of taste. They are not involved in nutrient absorption in the intestine.
- **Alveoli are tiny air sacs present in the lungs**. They provide a large surface area for the exchange of oxygen and carbon dioxide during respiration. Although they also increase surface area, they are part of the respiratory system, not the digestive system.

Q 101. Ans. C

Explanation:

- The India AI Impact Summit 2026 was held from February 16 to February 21, 2026 at Bharat Mandapam in New Delhi, and was inaugurated by PM Narendra Modi. The central theme of the summit was "**Sarvajan Hitaya, Sarvajan Sukhaya**" (Welfare for All, Happiness for All), reflecting India's vision that AI should serve humanity and contribute to inclusive growth.
- A major highlight of the summit was the presentation of India's **M.A.N.A.V. framework**, which outlines the country's approach to responsible and human-centric AI governance. Under this framework, **M.A.N.A.V.** stands for **Moral and Ethical Systems, Accountable Governance, National Sovereignty, Accessible and Inclusive, and Valid and Legitimate**. The framework emphasizes that AI systems should be developed and deployed in a manner that is ethical, transparent, accountable, inclusive, and aligned with national interests. It seeks to ensure that technological progress benefits society while safeguarding human rights, privacy, and public trust.

Q 102. Ans. D

Explanation:

- **Statement 1 is correct:** Mission MITRA (Mapping of Interoperable Traits and Response Assessment) is a first-of-its-kind **team behavioural study** developed by ISRO specifically for human spaceflight missions under the **Gaganyaan programme**. It evaluates crew safety and performance in human spaceflight by studying how astronauts function as a team under extreme stress.
- **Statement 2 is correct:** Mission MITRA is **jointly designed and conducted** by ISRO and the **Indian Air Force's Institute of Aerospace Medicine (IAM)**. The Human Space Flight Centre (HSFC) of ISRO leads the mission, while the IAF-Institute of Aerospace Medicine provides critical medical and behavioural expertise.
- **Statement 3 is correct:** The primary objective of Mission MITRA is to examine the **physiological, psychological, and operational dynamics** of Gaganyaan crew and ground teams functioning in high-altitude environments. The study specifically evaluates stress resilience, team cohesion, decision-making under pressure, interpersonal cohesion, and emotional stability—all critical for long-duration space missions. The test was conducted at **Leh, Ladakh** (approximately 3,500–4,000 meters above sea level), simulating extreme conditions like **hypoxia, low temperature, and isolation** to replicate space-like environments

Q 103. Ans. B

Explanation:

- **Statement 1 is incorrect:** the SHANTI Act received the assent of the President of India on **20 December 2025** and was gazette notified the same day. So the date mentioned in the question, **20 October 2025**, is wrong.
- **Statement 2 is correct:** SHANTI Act provides a legal framework for the promotion, development and use of atomic energy and ionising radiation for public welfare, including applications in healthcare, food, water, agriculture, research, industry, environment, and nuclear science and technology. This makes it a broad governing framework for India's civil nuclear sector, not just for power generation.
- **Statement 3 is correct:** Act permits private companies to participate in India's nuclear sector, including plant operations and power generation, subject to licensing and safety authorisation. The framework therefore opens the sector to private

participation, while regulated and safety-sensitive activities remain under government oversight.

Q 104. Ans. B

Explanation:

- **Zinc Finger Nucleases (ZFNs)** are engineered **gene-editing tools used to make targeted cuts at specific locations in DNA**. They consist of **two components**: a **zinc-finger protein domain**, which recognizes and binds to a specific DNA sequence, and a **nuclease domain (usually FokI)**, which cuts the DNA. Once the DNA is cut, the cell's natural repair mechanisms can be used to insert, delete, or modify genetic material. Thus, the primary function of ZFNs is **precise DNA cutting for genome editing**.

Q 105. Ans. B

Explanation:

- The **Bletchley Declaration** was adopted during the **AI Safety Summit 2023** held at Bletchley Park in the **United Kingdom**. It was signed by **28 countries and the European Union** with the objective of establishing a common understanding of the opportunities and risks associated with **frontier Artificial Intelligence (AI)**. Frontier AI refers to highly capable and advanced AI systems that may have transformative impacts on economies, societies, national security, and scientific research. The declaration emphasizes international cooperation to identify, assess, and manage AI-related risks while harnessing AI's benefits for humanity.

Q 106. Ans. D

Explanation:

- The international art exhibition titled "**Topographies of Tents, Terracotta, and Time**" will be hosted in Hyderabad. The event, a collaboration between Srishti Art Gallery and Goethe-Zentrum Hyderabad, will showcase the works of renowned international and national artists, focusing on themes such as identity, displacement, and transformation. This exhibition offers an immersive experience, with pieces from artists like Arnaldo Dres Gonzalez, Sudhakar Chippa, and Swen Kahlert, who bring their unique perspectives on migration, urbanization, and nature.

Q 107. Ans. C

Explanation:

- India recently assumed the chairmanship of the **Bay of Bengal Programme Inter-Governmental**

Organisation (BOBP-IGO) from Bangladesh during the organisation's Governing Council meeting.

- **The BOBP-IGO is a regional fisheries body that promotes:**

- ✓ Sustainable management of marine fisheries resources
- ✓ Livelihood security of small-scale fishers
- ✓ Regional cooperation among countries bordering the Bay of Bengal

- Member countries include Bangladesh, India, Maldives and Sri Lanka.

Q 108. Ans. C

Explanation:

- Bitra Island is the smallest inhabited island of Lakshadweep and is located close to Agatti Island in the Arabian Sea. The island occupies a strategically important position in the Lakshadweep archipelago and lies near important maritime routes.

- **Additional Facts:**

- ✓ Bitra has a very small population and limited land area.
- ✓ It is part of the Aminidivi group of islands in Lakshadweep.
- ✓ Agatti Island is one of the most significant islands of Lakshadweep as it hosts the only operational airport in the Union Territory.

Q 109. Ans. B

Explanation:

- At the National Anubhav Awards 2025, State Bank of India (SBI) and Bharat Heavy Electricals Limited (BHEL) emerged as first-time awardees. The awards recognize retiring employees and pensioners for documenting their work experiences and administrative insights, thereby preserving valuable institutional knowledge.

- **Additional Facts:**

- ✓ The Anubhav Awards Scheme is administered by the Department of Pension & Pensioners' Welfare (DoPPW).
- ✓ It encourages retiring government employees to share experiences and best practices from their service.
- ✓ The initiative promotes knowledge sharing and helps improve governance through documented institutional memory.

Q 110. Ans. A

Explanation:

- India's first Artificial Intelligence (AI)-enabled

Anganwadi has been established at Waddhamna village in Nagpur district, Maharashtra. The initiative integrates AI-based tools to improve early childhood care, nutrition monitoring, growth assessment, and pre-school education services delivered through the Anganwadi system.

- **Additional Facts:**

- ✓ The project aims to enhance the effectiveness of the Integrated Child Development Services (ICDS) framework.
- ✓ AI-based systems assist in tracking children's growth parameters and developmental milestones.

Q 111. Ans. C

Explanation:

- **The Rise of the Hitman — R. Kaushik**

- ✓ This sports biography traces the journey of the Indian cricket team captain Rohit Sharma (affectionately known as the "Hitman"). It chronicles his rise from his early days in Borivali to leading India to international glory, including the T20 World Cup win.

- **M.S. Swaminathan: The Man Who Fed India — Priyambada Jayakumar**

- ✓ The book chronicles the life and contributions of M. S. Swaminathan, widely regarded as the architect of India's Green Revolution.

- **The Man Who Became Cinema: Dilip Kumar — Ashok Chopra**

- ✓ The book is authored by Ashok Chopra and examines the life, career, and enduring legacy of legendary actor Dilip Kumar, often referred to as the "Tragedy King" of Indian cinema.

Q 112. Ans. C

Explanation:

- **Statement 1 is correct:** PSLV-C62/EOS-N1 is the 9th dedicated commercial mission undertaken by NewSpace India Limited (NSIL). It carries the EOS-N1 earth observation satellite along with 15 co-passenger satellites from domestic and international customers.

- **Statement 2 is correct:** This PSLV-C62/EOS-N1 mission marks the 64th flight of PSLV and uses the PSLV-DL variant, which is characterised by two solid strap-on motors attached to the first stage for additional thrust during launch.

Q 113. Ans. C

Explanation:

- The Eastern Imperial Eagle (*Aquila heliaca*),

recently sighted in Mudumalai Tiger Reserve, is listed as Vulnerable on the IUCN Red List.

● **About the Eastern Imperial Eagle:**

- ✓ It is a large migratory bird of prey found across parts of Europe, Central Asia, and South Asia.
- ✓ India serves as an important wintering ground for the species.
- ✓ The eagle inhabits open woodlands, grasslands, and semi-arid landscapes.
- ✓ Major threats include habitat loss, electrocution from power lines, poisoning, and decline in prey populations.
- ✓ It is listed under Appendix I and II of CITES.
- ✓ The species is protected under the Convention on Migratory Species (CMS).

Q 114. Ans. C

Explanation:

- The Defence Research and Development Organisation (DRDO) recently successfully conducted field evaluation trials of the third-generation Man Portable Anti-Tank Guided Missile (MPATGM) at the KK Ranges, Ahilya Nagar (formerly Ahmednagar), Maharashtra.
- **About MPATGM:**
 - ✓ It is an indigenously developed, fire-and-forget anti-tank guided missile.
 - ✓ Designed for deployment by infantry units against modern armoured vehicles.
 - ✓ Equipped with an Imaging Infrared (IIR) Seeker for enhanced target acquisition and accuracy.
 - ✓ The missile is intended to strengthen the Indian Armed Forces' indigenous anti-tank warfare capabilities.

Q 115. Ans. A

Explanation:

- The world's longest expressway tunnel was recently inaugurated in China's Xinjiang Uyghur Autonomous Region. The tunnel is a key component of the country's transport infrastructure and significantly improves connectivity across the rugged Tianshan Mountains, reducing travel time and enhancing economic integration in the region.
- **Additional Fact:**
 - ✓ The tunnel forms part of the Urumqi-Yuli Expressway project.
 - ✓ It is designed to facilitate year-round transportation by providing a safer and faster route through the mountainous terrain.

- ✓ Xinjiang is China's largest provincial-level administrative region and plays a strategic role in the Belt and Road Initiative (BRI).

Q 116. Ans. C

Explanation:

- The 11th Global Forum of the United Nations Alliance of Civilizations (UNAOC) was hosted in **Riyadh**, the capital city of Saudi Arabia, from 14 to 15 December 2025.
- **The forum was held under the central theme "UNAOC: Two Decades of Dialogue for Humanity, Advancing a New Era of Mutual Respect and Understanding in a Multipolar World."**
- Marking the 20th anniversary of the UNAOC, the high-level international gathering brought together state leaders, ministers, religious figures, civil society representatives, and youth from around the globe. The event focused on fostering intercultural and interreligious dialogue, combating polarisation, and building global peace, culminating in the adoption of the Riyadh Declaration.

Q 117. Ans. B

Explanation:

- The EARTH Summit 2025 (Empowering Agriculture, Rural Transformation & Harmony) was organised jointly by NABARD and IAMAI (Internet and Mobile Association of India) and was held at the Mahatma Mandir Convention & Exhibition Centre, Gandhinagar, Gujarat. The summit focused on rural innovation, cooperative-led development, AgriTech, RuralTech, and strengthening the rural economy. . Governance Index, Shiksha Sarathi, and Grain Storage Application were launched.

Q 118. Ans. A

Explanation:

- **Statement 1 is incorrect:** The Impact RISE (Research, Innovation, Skilling and Entrepreneurship) Initiative was launched by Indian Institute of Technology Kharagpur, not by IIT Bombay. The initiative aims to drive large-scale and measurable societal transformation through sustainable technology, innovation, skilling and entrepreneurship.
- **Statement 2 is correct:** The initiative focuses on several impact-oriented sectors, including women's health, clean water technologies, climate resilience, AI-based health diagnostics, and rural micro-grids/clean energy solutions

Q 119. Ans. C

Explanation:

- Vikram-I is India's first privately developed orbital-class rocket, unveiled by Skyroot Aerospace in Hyderabad. It is designed to place small satellites into Low Earth Orbit (LEO) and represents a major milestone in India's private space sector.
- **Additional Facts:**
 - ✓ Vikram-S was a suborbital launch vehicle and became the first private Indian rocket to be successfully launched in 2022 under the Prarambh mission.
 - ✓ Vikram-I is a four-stage launch vehicle (comprising three solid-propellant stages and one liquid-propellant upper stage) capable of carrying satellites weighing up to around 480 kg to low Earth orbit.

Q 120. Ans. C

Explanation:

- Hyderabad-based space-tech startup Stardour successfully test-fired India's first indigenous hydrogen-oxygen propulsion engine, marking a major milestone in the country's private space sector. The engine is intended to power "Lucas," Stardour's next-generation Orbital Transfer Vehicle (OTV) for deep-space and orbital logistics missions. The test was conducted at the Indian Institute of Science (IISc), Bengaluru.

Q 121. Ans. C

Explanation:

- **Pidiya Painting (also spelled Pidhiya/Pidiha)** is a traditional folk art form of the **Bhojpur region of Bihar** and the broader Bhojpuri cultural area. Deeply rooted in local customs and family traditions, it is especially associated with **Govardhan Puja** and symbolizes the enduring bond between brothers and sisters. The paintings are characterized by **geometric designs, vibrant motifs, and depictions of rural life**, including scenes of farming, village fairs, animals, and birds. Reflecting the cultural heritage of the Bhojpur region, the art form has been preserved through generations.
- Recognizing its unique cultural significance, **Pidiya Painting has been granted a Geographical Indication (GI) tag**. It received the certification along with **Bawan Buti Saree** of Nalanda and **Pattharkatti Stone Craft** of Gaya, two other renowned traditional crafts of Bihar.

Q 122. Ans. A

Explanation:

- **Kusiargaon Biodiversity Park**, located in **Araria district**, is **Bihar's first and largest biodiversity park, spread over more than 200 acres**. Developed as an ecological, recreational, and research hub, the park plays a significant role in biodiversity conservation and environmental awareness. It features a large herbal garden, butterfly enclosure, climbers' garden, eco-huts, and a Nature Interpretation Centre. Situated at Kusiargaon, about 10 km from the Araria district headquarters on NH-57, the park has emerged as an important destination for nature education and eco-tourism in Bihar.
- The **RPCAU Biodiversity Park**, situated within the campus of **Dr. Rajendra Prasad Central Agricultural University** at Pusa in **Samastipur district**, was established to conserve native flora and fauna. Spread over 7.5 hectares, the park serves as a center for biodiversity conservation, environmental education, and research. It functions as an outdoor learning facility where students, researchers, and visitors can gain practical knowledge about ecological preservation and sustainable environmental practices.

Q 123. Ans. D

Explanation:

- **Pair 1 is correct: Kusheshwar Asthan Lake**, located in **Darbhangha district**, is one of Bihar's most important wetlands and extends across nearly 14 villages. The wetland is named after the nearby **Baba Kusheshwar Nath Temple**, a revered Lord Shiva temple often called the "Baba Dham of Mithila." Locally known as a *chaur*, the wetland is formed by the overflow of the **Kamla** and **Kareh** rivers, creating a rich ecosystem of marshes, water hyacinths, and native grasses.
- **Pair 2 is correct: Jagatpur Lake**, situated in **Bhagalpur district**, is an important wintering ground for migratory birds and waterfowl. The lake is particularly known as one of the key habitats of the vulnerable **Greater Adjutant Stork**. Other notable bird species found here include the **Oriental Darter**, **Black-necked Stork**, and **Cotton Pygmy Goose**, making it a significant site for bird conservation in Bihar.
- **Pair 3 is correct: Anupam Lake**, located within the **Kaimur Wildlife Sanctuary**, is a picturesque natural water body surrounded by forests and wildlife

habitats. The lake is a popular destination for nature lovers and wildlife enthusiasts and contributes to the ecological richness of the sanctuary.

- **Pair 4 is correct: Muchalinda Lake**, situated in **Bodh Gaya**, holds immense religious and historical significance in Buddhism. At the center of the lake stands a meditative statue of Lord Buddha. According to Buddhist tradition, during the sixth week after attaining enlightenment, Buddha was meditating when the region experienced a severe seven-day storm. The serpent king **Muchalinda** emerged from the waters, coiled around Buddha, and spread his hood above him to protect him from the rain and wind, allowing him to continue his meditation undisturbed. This episode is commemorated through the lake and its iconic sculpture.

Q 124. Ans. A

Explanation:

- Bihar's hot springs are mainly concentrated in two regions—**Rajgir in Nalanda district** and **Munger district**—and are associated with natural geothermal activity.
- The **Rajgir Hot Springs**, located at the foothills of the Vaibhava Hills in **Nalanda district**, comprise a group of thermal springs known as the **Saptadhara**. Among them, **Brahmakund** is the hottest and most sacred spring, while **Suryakund** is another important thermal spring associated with Sun worship and revered in Hindu, Buddhist, and Jain traditions. Therefore, **Statements 1 and 2 are correct**.
- On the other hand, **Sita Kund, Rishi Kund, and Bhimbandh** are located in the **Munger region**. Sita Kund is known for its religious significance, Rishi Kund is situated in the forested Kharagpur Valley, and Bhimbandh is famous for its warm water sources within the Bhimbandh Wildlife Sanctuary. Hence, **Statements 3, 4, and 5 are incorrect**.

Q 125. Ans. C

Explanation:

- **Kakolat Waterfall**, located in **Nawada district** near the Bihar–Jharkhand border, is one of the most famous natural attractions in Bihar. Cascading from a height of about **160 feet**, the waterfall is renowned for its crystal-clear, cold waters and the natural pool formed at its base, making it a popular destination during the summer season.
- Owing to its scenic beauty, lush green surroundings, and impressive cascade, Kakolat Waterfall is often

referred to as the “**Niagara of Bihar**” or the “**Mini Kashmir of Bihar**.” The site also holds religious and historical significance in local traditions. According to Hindu folklore, the **Pandavas** are believed to have visited the area during their exile, while another legend associates the waterfall with **Lord Krishna**, who is said to have bathed here along with his queens.

Q 126. Ans. C

Explanation:

- **Assertion (A) is true:** In Bihar, the **rainy season generally begins in mid-June** with the arrival of the **Southwest Monsoon** and continues until around September or early October. Most of the state's annual rainfall is received during this period.
- **Reason (R) is false:** The onset of the monsoon in Bihar is **not caused by a storm from the Arabian Sea passing over the state**. Bihar receives the bulk of its rainfall from the **Bay of Bengal branch of the Southwest Monsoon**, which advances northwestward across eastern India and reaches Bihar around the middle of June. Although the Arabian Sea branch contributes to India's monsoon system, it is not the primary reason for the onset of monsoon in Bihar.

Q 127. Ans. A

Explanation:

- Bihar's **GSDP at constant prices** is estimated to grow by **8.6% in 2024–25** over the previous year.
- In 2024–25, the **sector-wise contribution to Bihar's economy (at current prices)** is:
 - ✓ **Agriculture:** 23%
 - ✓ **Manufacturing:** 23%
 - ✓ **Services:** 54%
- The **services sector** is the largest contributor to Bihar's economy, accounting for more than half of the GSDP.
- Bihar's **per capita GSDP** is estimated at **₹76,490 in 2024–25**, showing a **13% increase** over the previous year.
- India's per capita GDP for 2024–25 is estimated at **₹2,34,859**, an increase of 9% over the previous year.
- Bihar's projected **GSDP for 2026–27 (at current prices)** is **₹13.1 lakh crore**, reflecting a growth of **15% over 2025–26**.
- The budget estimates a **revenue surplus of ₹1,143 crore (0.1% of GSDP)** in 2026–27.
- Bihar has targeted a **fiscal deficit of 3% of GSDP** in 2026–27.

Q 128. Ans. C

Explanation:

- The **Bihar AI Summit 2026**, held on **23–24 May 2026** at the **Urja Auditorium, Patna**, was the state's flagship event dedicated to **artificial intelligence and digital innovation**. The summit brought together **policymakers, industry experts, startups, academicians, and technology leaders** with the objective of accelerating Bihar's digital transformation and positioning the state as an emerging centre for AI-driven development. A key focus of the discussions was the creation of AI applications in **regional languages such as Hindi, Bhojpuri, and Maithili**, enabling wider adoption and accessibility at the grassroots level. The event also served as a showcase for innovation, providing a platform for **more than 100 startups** to exhibit their ideas, conduct **live robotic demonstrations**, and engage with potential investors and industry stakeholders.

Q 129. Ans. A

Explanation:

- The **Jai Prakash Narayan Bird Sanctuary**, popularly known as **Surha Taal**, is situated in the **Ballia district of Uttar Pradesh**. Located approximately **10–17 km from Ballia city**, the sanctuary is centered around **Surha Taal**, a natural **oxbow lake** formed by an abandoned meander of the **Ganga River**. The wetland is an important habitat for a variety of resident and migratory birds. Recognizing its ecological significance, it was designated as **India's 100th Ramsar Site (Wetland of International Importance)** under the **Ramsar Convention**.

Q 130. Ans. C

Explanation:

- The **Bihar Panchayati Raj Act, 2006**, introduced by the Nitish Kumar government, brought significant changes in local self-governance by expanding reservations in Panchayati Raj Institutions.
- **Key provisions included:**
 - ✓ **50% reservation for women** in all three tiers of Panchayati Raj institutions.
 - ✓ **20% reservation for Extremely Backward Classes (EBCs)**.
 - ✓ **Reservation for Scheduled Castes (SCs) and Scheduled Tribes (STs)** in proportion to their population.

Q 131. Ans. D

Explanation:

- The **Ajay River** originates near **Chakai and Batpar** in the **Jamui district of Bihar**. Within Bihar, the river remains largely in its **upper (youthful) mountainous course**, where it is mainly fed by **seasonal streams and local rivulets**. It does not receive any major named perennial tributaries before entering Jharkhand.
- **Most of the river's significant tributaries join it further downstream in Jharkhand and West Bengal. These include:**
 - ✓ **Pathro River** – a major right-bank tributary originating in the **Deoghar and Giridih districts of Jharkhand**.
 - ✓ **Jayanti River** – another important right-bank tributary that rises in the forested regions of **Jharkhand**.
 - ✓ **Kunur River** – a major right-bank tributary that joins the Ajay in **Purba Bardhaman district, West Bengal**.
 - ✓ **Tumuni River** – a smaller tributary flowing through the **Bardhaman region of West Bengal**.
- Thus, while the Ajay River begins in Bihar, its principal tributary network develops mainly during its course through **Jharkhand and West Bengal**.

Q 132. Ans. D

Explanation:

- **Statement 1 is correct:** Under the **Bihar Panchayat Raj Act, 2006**, a **Gram Katchahry** has been established at every **Gram Panchayat** to provide justice at the grassroots level and facilitate settlement of minor civil and criminal disputes within villages. The Panchayati Raj Department of Bihar explicitly states that Gram Katchahrys were established at each Gram Panchayat for "dispensation of justice at the doorsteps of rural populace."
- **Statement 2 is correct:** The Gram Katchahry has limited judicial powers. After hearing the parties and examining evidence, it may impose a **fine not exceeding ₹1,000** on a person found guilty. It cannot award imprisonment. This power is provided under **Section 107 of the Bihar Panchayat Raj Act, 2006**.

Q 133. Ans. B

Explanation:

- **Patna Kalam**, also known as the **Patna School of Painting**, flourished in Bihar during the 18th and

19th centuries. It was developed by artists trained in the Mughal tradition and is officially described by the Bihar Tourism Department as a painting style that **blends Mughal and Persian art styles**. The school became famous for depicting the daily lives of ordinary people rather than royal courts and aristocratic themes.

Q 134. Ans. B

Explanation:

- The Munger Hilly region (which includes the Kharagpur Hills) is located in the southern fringe of Bihar and forms part of the state's Southern Plateau and Hilly Region. This region consists of scattered hill tracts that mark the geographical boundary of Bihar. While the Kaimur hills in the southwest are an extension of the Vindhyan range, the hills of Rajgir, Gaya, and Munger are northern extensions of the ancient Chota Nagpur Plateau system (featuring Dharwar rock formations).

Q 135. Ans. D

Explanation:

- **Statement 1 is correct:** Khadar (Newer Alluvium) soil is formed by the continuous and recent deposition of sediments brought down by the Himalayan and plains rivers. In Bihar, it is primarily spread across the floodplains and lower valleys of the Ganga, Gandak, Burhi Gandak, Kosi, and Mahananda rivers.
- **Statement 2 is correct:** Because this soil is renewed annually by floods, it is extensively distributed across the low-lying northern and eastern floodplain districts of the state. Prominent districts include Darbhanga, Saharsa, Supaul, Bhagalpur, Katihar, Purnia, and Muzaffarpur.
- **Statement 3 is correct:** Chemically and physically, Khadar soil is generally dark brown to light brown in colour and has a predominant mix of sand, silt, and clay. While it is naturally deficient in nitrogen and humus, the regular top-up of fresh river alluvium makes it highly fertile and excellent for water-intensive crops like paddy and jute.

Q 136. Ans. B

Explanation:

- **Assertion (A) is true** Most of the major rivers flowing through northern Bihar, such as the Ghaghara, Gandak, Burhi Gandak, Bagmati, Kamla, Kosi, and Mahananda, originate in the Himalayan or Trans-Himalayan regions. These

rivers carry enormous quantities of silt, sand, and other sediments eroded from the young and fragile Himalayan mountains. As a result, they frequently change their courses and contribute to flooding and alluvial soil formation in northern Bihar.

- **Reason (R) is true:** The major rivers of southern Bihar, such as the Son, Punpun, Kiul, Badua, Chanan, and Sakri, generally originate in the Chhotanagpur Plateau and nearby uplands. Their flow depends largely on seasonal rainfall, making them less perennial and more variable in discharge than the Himalayan rivers.
- However, **(R) does not explain (A)**. The fact that southern rivers originate in the Chhotanagpur Plateau and are rainfall-fed does not explain why northern rivers originate in the Himalayas and carry heavy sediment loads. The two statements describe different characteristics of Bihar's river systems.

Q 137. Ans. B

Explanation:

The **Smart Cities Mission**, launched on **25 June 2015**, aims to improve the quality of life in selected cities by providing core infrastructure, sustainable urban development, efficient governance, and smart technological solutions.

- **In Bihar, four cities were selected under the mission:**
 - ✓ Patna
 - ✓ Bhagalpur
 - ✓ Bihar Sharif
 - ✓ Muzaffarpur
- Therefore, among the cities listed in the question, **Patna (1), Bhagalpur (2), and Muzaffarpur (4)** are part of the Smart Cities Mission, while **Gaya (3)** was **not selected** under the mission.
- **Patna** focuses on modernization of civic amenities, intelligent traffic management, improved water supply, and redevelopment of public spaces.
- **Bhagalpur**, the first city from Bihar selected under the mission, emphasizes smart roads, solar energy integration, and sanitation improvements.
- **Muzaffarpur** prioritizes urban mobility, traffic decongestion, solid waste management, and socioeconomic development.
- **Bihar Sharif** concentrates on green spaces and rejuvenation of water bodies such as the Dhaneshwari Ghat pond.

Q 138. Ans. C

Explanation:

Bihar's expenditure in 2026-27 on six key sectors as a

proportion of its total expenditure on all sectors. The average for a sector indicates the average expenditure in that sector by 31 states and UTs (including Bihar) as per their budget estimates of 2025-26.

- **Education:** Bihar has allocated 21.7% of its expenditure on education in 2026-27. This is significantly higher than the average allocation for education by states in 2025-26 (14.5%).
- **Health:** Bihar has allocated 6.3% of its expenditure on health in 2026-27. This is marginally higher than the average allocation for health by states in 2025-26 (6.2%).
- **Rural development:** Bihar has allocated 9.4% of its expenditure on rural development in 2026-27. This is significantly higher than the average allocation for rural development by states in 2025-26 (4.9%).
- **Roads and bridges:** Bihar has allocated 2.9% of its expenditure on roads and bridges in 2026-27. This is lower than the average allocation for roads and bridges by states in 2025-26 (4.3%).
- **Agriculture:** Bihar has allocated 2.6% of its expenditure on agriculture in 2026-27. This is significantly lower than the average allocation for agriculture by states in 2025-26 (5.7%).
- **Police:** Bihar has allocated 5.2% of its expenditure on police in 2026-27. This is higher than the average allocation for police by states in 2025-26 (4.0%).

Q 139. Ans. B

Explanation:

- **Balthar Soil:** It is a transitional soil found in a narrow belt along the Chota Nagpur Plateau fringe (from Kaimur to Banka). Because it is formed by the weathering of ancient crystalline plateau rocks, it is rich in iron (giving it a reddish-yellow tint) but notoriously poor in lime (calcium carbonate) and humus, making it acidic in nature.
- **Balsundari:** This is an old alluvium soil found in North Bihar (Saran, Muzaffarpur, Darbhanga). Unlike Balthar, it is highly alkaline/calcareous and exceptionally rich in lime.
- **Karail-Kewal:** This is the old alluvium soil of the South Bihar plains. It is characterized by heavy clay, a neutral-to-alkaline pH, and high water-retention capacity, completely opposite to the sandy, acidic nature of Balthar.
- **Tal Soil:** This is found in backwater patches south of the Ganga River. It is a heavy, dark, and highly fertile clayey soil that remains submerged during the monsoons.

Q 140. Ans. B

Explanation:

- **Assertion (A) is true:** Bihar's climate is occasionally influenced by **typhoons and tropical disturbances originating in the South China Sea**. After crossing Southeast Asia and Myanmar, some of these systems weaken and merge with weather disturbances over the Bay of Bengal, ultimately affecting eastern and northern India, including Bihar.
- **Reason (R) is true:** During the **retreating monsoon season (October–November)**, Bihar can experience rainfall and weather disturbances caused by **tropical cyclones originating in the Bay of Bengal**. These cyclonic systems often move inland and influence the state's weather.
- However, (R) **does not explain (A)**. The influence of South China Sea typhoons on Bihar and the impact of Bay of Bengal cyclones are two separate meteorological phenomena. While both affect Bihar's climate, the occurrence of Bay of Bengal cyclones is not the reason why Bihar is influenced by typhoons originating in the South China Sea.

Q 141. Ans. A

Explanation:

Green	Plant	Water	→	18	42	63
	Water	Soil	→	63	27	81
	Plant	Crop	→	42	81	54

Q 142. Ans. C

Explanation:

Observe the relation between the letters and the number.

Term	Letter Positions	Sum
BF8	B=2, F=6	2 + 6 = 8
DH12	D=4, H=8	4 + 8 = 12
JL20	J=10, L=12	10 + 12 = 22
RV40	R=18, V=22	18 + 22 = 40

- In BF8, DH12 and RV40, the number is equal to the sum of the alphabetical positions of the two letters. However, for JL20, the sum of the positions of J and L is 22, not 20.
- Therefore, JL20 is the odd one out.

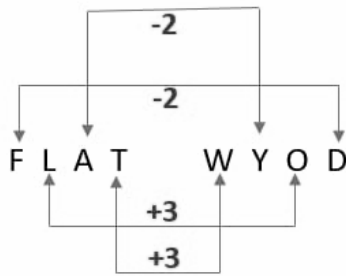
Q 143. Ans. B

Explanation:

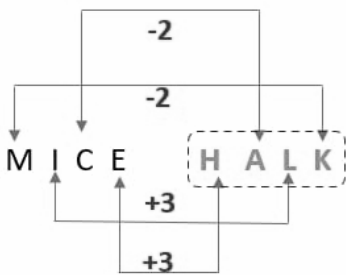
The logic followed here is 1st and 3rd letters are replaced by their 2nd preceding letters, 2nd and 4th

letters are replaced by 3rd succeeding letters as per English alphabetical series. And then all the letters are written in reverse order to form the coded word.

In FLAT,



Similarly, In MICE



Q 144. Ans. A

Explanation:

- Count the triangles systematically according to their sizes.
- Small triangles = 10
- Medium triangles = 3
- Large triangles = 1
- Total triangles = $10 + 3 + 1 = 14$
- Therefore, the number of triangles in the figure is 14.

Q 145. Ans. C

Explanation:

- The logic followed here is, the square of the third number is subtracted from the sum of squares of first two numbers from the left end.
- In $5^2 \times 8^2 = 85$
 $\checkmark (5^2 + 8^2) - 2^2 = (25 + 64) - 4 = 85$
- In $6^2 \times 9^2 = 108$
 $\checkmark (6^2 + 9^2) - 3^2 = (36 + 81) - 9 = 108$
- Similarly, in $10^2 \times 4^2 = ?$
 $\checkmark (10^2 + 4^2) - 7^2 = (100 + 16) - 49 = 67$

Q 146. Ans. A

Explanation:

- Since the two numbers are in the ratio 7:5, they can be represented as $7k$ and $5k$ (where k is an integer).
 $\checkmark$ HCF = k
 $\checkmark$ LCM = $35k$
- Given that:

- $\checkmark$ HCF $\times$ LCM = 875
- $\checkmark$ $k \times 35k = 875$
- $\checkmark$ $35k^2 = 875$
- $\checkmark$ $k^2 = 25$
- $\checkmark$ $k = 5$

- Therefore, HCF = 5
 $\checkmark$ LCM = $35 \times 5 = 175$
- Sum of their reciprocals = $1/175 +$
 $\checkmark = 1/175 + 35/175$
 $\checkmark = 36/175$

Q 147. Ans. A

Explanation:

We know that,
 $a^2 - b^2 = (a+b)(a-b)$
 $\Rightarrow (31.5)^2 - (18.5)^2$
 $= (31.5 + 18.5)(31.5 - 18.5)$
 $\Rightarrow 50 \times 13$
 $\Rightarrow 650$
 Therefore, the required value is 650.

Q 148. Ans. C

Explanation:

Value of $\sqrt{p^{-1}q} \times \sqrt{q^{-1}r} \times \sqrt{r^{-1}p}$
 $\Rightarrow \sqrt{\frac{1}{p} \times q \times \frac{1}{q} \times r \times \frac{1}{r} \times p}$
 $\Rightarrow \sqrt{1} = 1$

Q 149. Ans. D

Explanation:

- P's work rate = Total work / Day = $1/10$
- Since Q completes half the work in 15 days, he will complete the full work in 30 days.
- Q's rate = $1/30$
- Since R completes $1/3$ of the work in 10 days, he will complete the full work in 30 days.
- R's rate = $1/30$
- The combined rate of the three = $1/10 + 1/30 + 1/30 = 5/30 = 1/6$
- Therefore, the time required to complete the entire work = 6 days

Q 150. Ans. D

Explanation:

- Radius of the circle (r) = 14 cm
- Circumference of the circle = $2 \times 22/7 \times 14 = 88$ cm
- Circumference of the circle = Perimeter of the square
- Therefore, Perimeter of the square = 88 cm
- Now, Side of the square = $88 / 4 = 22$ cm
- Area of the square = $22^2 = 484$ cm²

73rd BPSC Foundation Batch

Prelims + Mains + Interview

BATCH 9

हिंदी माध्यम

📍 Musallahpur

ONLINE & OFFLINE

STARTS FROM 9th APRIL 2026

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026

BATCH 6

Bilingual

BATCH 1

English Medium

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026



Patna

Musallahpur

KGS Campus,
Near Sai Mandir,
Musallahpur Hat,
Patna, Bihar-800006

Boring Road

2nd Floor, Kumar
Tower, Boring Road
Chauraha, Patna,
Bihar-800001

Naya Tola

KP Yadav
Compound Kazipur,
Naya Tola, Patna,
Bihar-800004